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ORIGINAL PAPER

Explainable and physics-informed machine learning for seasonal water quality prediction in the monsoon-driven Padma River Basin, Bangladesh

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Abstract River water quality in monsoon-driven subtropical basins exhibits strong seasonal variability driven by hydroclimatic forcing and increasing anthropogenic pressure, posing challenges for reliable assessment and management. Despite advances in water quality modeling, most Water Quality Index (WQI) prediction frameworks require extensive sampling and lack interpretability, limiting rapid baseline assessment during critical periods. This study develops the first integrated Explainable Artificial Intelligence (XAI) framework combining Machine Learning (ML), Deep Learning (DL), and Physics-Informed Neural Networks (PINNs) to predict, interpret, and spatially characterize seasonal water quality dynamics in the Padma River Basin, Bangladesh. Forty-four surface water samples collected during winter and monsoon seasons were evaluated using WQI assessment, explainable modeling, probabilistic uncertainty analysis, and spatial regionalization. Results show

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that seasonal variability dominates over spatial variability ($p < 0.0001$), with winter low-flow conditions promoting solute concentration and localized degradation, while monsoon discharge drives basin-wide dilution and recovery. Model performance is strongly region-dependent: Deep Neural Networks achieve the highest accuracy in winter ($R^2 = 0.98$; RMSE = 1.16), whereas Ridge Regression and Voting Ensemble models perform more robustly during the monsoon ($R^2 \approx 0.97$; RMSE ≈ 1.01). Explainable AI analysis identifies NO_3^- emerged as the dominant contaminant (24.0 ± 36.3 mg/L winter, 47.5 ± 68.7 mg/L monsoon, with isolated samples exceeding WHO limits), whereas pH and DO exhibit dual seasonal influences. PINN-based data augmentation improves model generalization under limited sampling while preserving hydrochemical consistency. Monte Carlo simulations quantify prediction uncertainty and reveal seasonal shifts in WQI probability distributions, while spatial autocorrelation analysis identifies localized winter degradation hotspots and widespread monsoon improvement. The proposed physics-informed and explainable AI framework enhances predictive reliability, interpretability, and decision relevance, offering a transferable approach for uncertainty-aware water quality assessment and adaptive management in monsoon-affected, data-limited river basins.

Keywords Artificial intelligence · DNN · Water Quality Index · Monsoon River Basin · Padma River · Bangladesh

Introduction

River water quality degradation has emerged as a critical global environmental challenge, particularly in rapidly developing regions where industrial expansion, urbanization, and population growth have outpaced wastewater treatment capacity and regulatory enforcement (Jawad-ul-Haque et al., 2023; Nishat et al., 2025). Recent international research increasingly emphasizes that river degradation is not governed solely by pollutant loading but is strongly modulated by hydroclimatic variability, seasonal flushing, and nonlinear interactions among physicochemical parameters, which together control ecological and human-health risks (Deng et al., 2022; Lakshmi, 2024; Paneru & Paneru, 2025; Zheng et al., 2025). For example, Bishwakarma et al. (2025) demonstrated pronounced spatiotemporal variability of trace elements and associated health risks in a major Himalayan River basin, highlighting how hydroclimatic drivers and heterogeneous pollution sources jointly shape river chemistry and basin-scale risk outcomes. Such findings underscore the growing consensus that static or single-season assessments are insufficient for diagnosing water quality dynamics in climatically sensitive river systems (Bishwakarma et al., 2025; Tapas et al., 2024).

In subtropical monsoon-dominated regions, these challenges are further amplified by sharp seasonal contrasts between low-flow winter conditions and high-discharge monsoon periods. Bangladesh exemplifies this vulnerability, where river systems play a central role in irrigation, fisheries, and domestic water supply while experiencing increasing anthropogenic pressure from agriculture, urban runoff, and industrial activity. The Padma River Basin is among the most socioeconomically significant systems in the country and has gained heightened environmental importance following the establishment of the Rooppur Nuclear Power Plant (RNPP), Bangladesh's first nuclear facility. This development necessitates robust baseline characterization and predictive capability for environmental risk assessment, emergency preparedness, and long-term monitoring (Islam et al., 2024; Uddin et al., 2017, 2018; Usha et al., 2025). Although several hydrochemical investigations have been conducted in the RNPP region, most were undertaken during the construction phase, leaving a critical gap in post-construction, seasonally resolved surface-water assessments that integrate prediction, uncertainty, and spatial diagnostics.

The Water Quality Index (WQI) remains one of the most widely used tools for synthesizing multiple physicochemical parameters into a single, communicable metric of overall water suitability (Bui et al., 2020; Horton, 1965). In Bangladesh, WQI-based evaluations are commonly benchmarked against national standards established by the Department of Environment (DoE, 2010). However, conventional WQI approaches are often location-dependent and insufficiently flexible to represent nonlinear parameter interactions and abrupt seasonal transitions characteristic of monsoon-driven river systems (Parasar & Krisna, 2025). As a result, traditional WQI calculations may obscure dominant pollution drivers,

underestimate uncertainty, and provide limited diagnostic value for adaptive management.

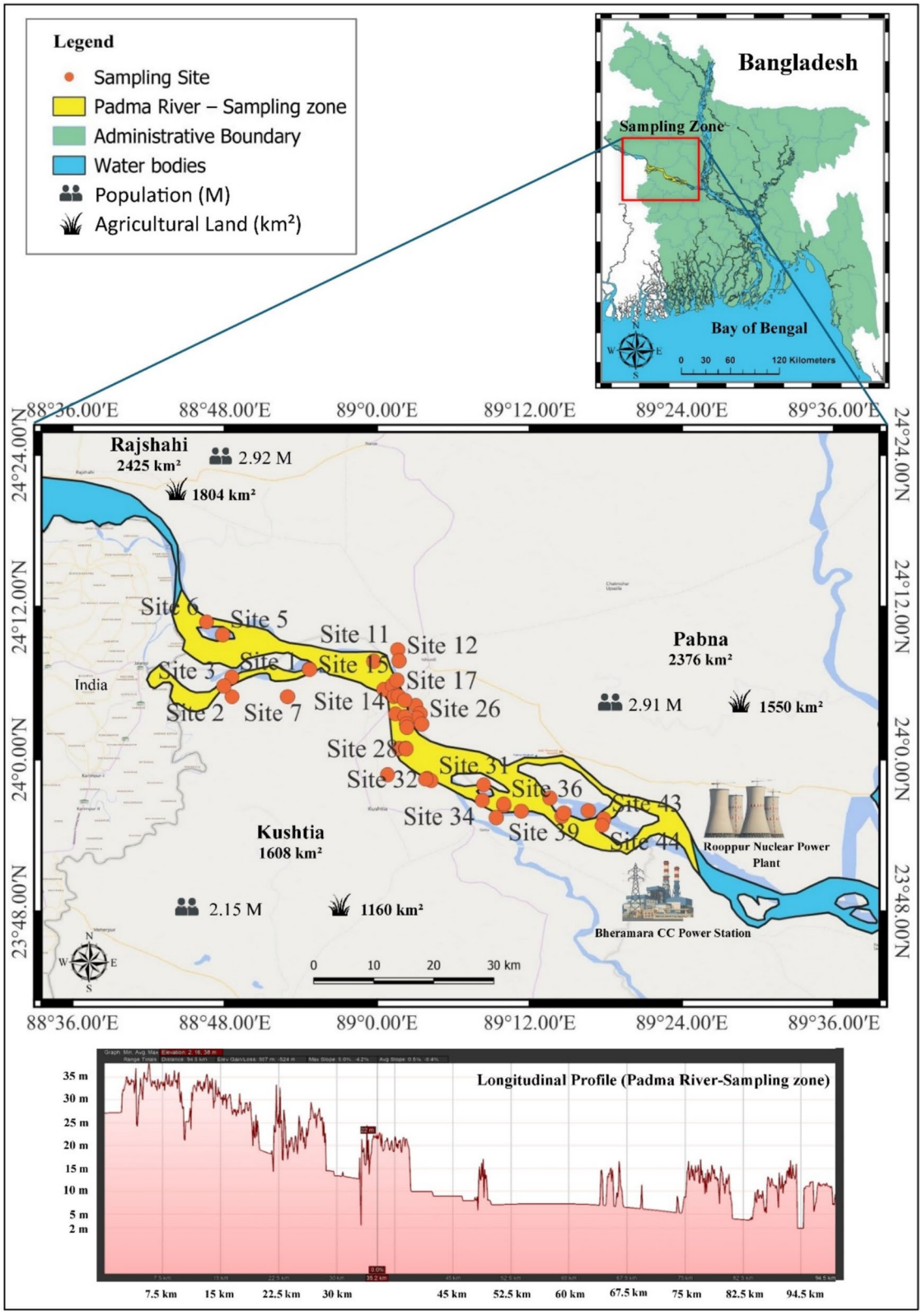
Physically based watershed and river models, such as SWAT and QUAL2K, have been extensively applied to simulate hydrological processes, sediment transport, and pollutant fate in river basins (Mummidivarapu et al., 2023). While these models offer valuable mechanistic insight, they require extensive calibration, dense input datasets, and detailed parameterization that are often impractical in data-scarce subtropical regions. Moreover, such process-based frameworks are not inherently designed to produce interpretable or probabilistic WQI estimates, nor to quantify the relative contribution of individual parameters to integrated water quality outcomes. Statistical trend analyses and fuzzy-logic approaches have therefore been adopted as computationally efficient alternatives (Islam et al., 2025), yet these methods typically rely on linear assumptions and struggle to capture complex, season-dependent hydrochemical interactions.

Recent advances in machine learning (ML) and deep learning (DL) have substantially improved predictive performance in river water quality modeling, particularly across monsoon-influenced river systems in Asia. A synthesis of representative ML- and DL-based river water quality studies, including model structures, predictor variables, and reported performance, is summarized in Supplementary Table S1, highlighting the rapid methodological evolution from conventional regression toward ensemble learning and deep neural architectures (Hui et al., 2025; Nair et al., 2024; Zheng et al., 2025). Tree-based ensembles, neural networks, and hybrid ML–DL frameworks frequently outperform conventional statistical models in short-term prediction and hotspot detection (Khosravi et al., 2025; Parasar & Krishna, 2025). However, many of these approaches function as "black boxes," limiting transparency and constraining their applicability in environmental decision-making, regulatory oversight, and risk communication (Imam et al., 2024; Li et al., 2024; Marshall et al., 2025).

Explainable Artificial Intelligence (XAI) has therefore emerged as a promising paradigm that combines predictive accuracy with interpretability through techniques such as SHAP and game-theoretic attribution (Satish et al., 2024; Zheng et al., 2025). XAI-based frameworks have improved understanding of variable importance in WQI prediction and enhanced stakeholder confidence in model outputs (Nong et al., 2025; Priyadharshini & Ramesh, 2025). Nevertheless, most existing XAI-enabled studies remain methodologically fragmented: they typically rely on observed datasets alone, lack explicit physical constraints to preserve hydrochemical realism under limited sampling, and rarely integrate uncertainty quantification or spatial dependence into a unified analytical framework (Kundu et al., 2025; Proshad et al., 2024). Consequently, predictive reliability across seasons and the spatial coherence of inferred degradation patterns remain insufficiently addressed.

To overcome these limitations, the present study proposes a conceptually integrated framework that explicitly links monsoon-driven hydroclimatic forcing with seasonal hydrochemical response, explainable ML/DL prediction, physics-informed data augmentation, probabilistic uncertainty characterization, and spatial regionalization. Physics-Informed Neural Networks (PINNs) are employed to augment limited seasonal datasets while enforcing hydrochemical consistency, thereby overcoming a key limitation of purely data-driven approaches (Raissi et al., 2017). Monte Carlo simulation is integrated to quantify uncertainty and characterize probabilistic WQI behavior, while spatial autocorrelation and network connectivity analyses are used to identify clustered degradation and recovery patterns across the basin (Proshad et al., 2024, 2025). SHAP-based XAI further enables transparent identification of season-specific pollution drivers and nonlinear parameter interactions.

The specific objectives of this study are to: (i) develop a hybrid Physics–ML–XAI framework for accurate seasonal WQI prediction; (ii) assess spatiotemporal variability of water quality under contrasting hydroclimatic conditions; (iii) interpret dominant physicochemical drivers and their interactions using DNN–SHAP analysis; and (iv) quantify uncertainty and perform spatial regionalization of WQI through Monte Carlo simulation and spatial autocorrelation. By integrating interpretability, physical consistency, uncertainty awareness, and spatial structure within a single workflow, this study advances beyond conventional WQI assessments, statistical trend analyses, and standalone process-based models. The proposed framework provides a transferable, decision-oriented approach for sustainable water quality management in monsoon-driven, data-limited river basins, with direct


Legend
Sampling Site
Padma River – Sampling zone
Administrative Boundary
Water bodies
Population (M)
Agricultural Land (km²)
Bangladesh
Sampling Zone
Bay of Bengal
0 30 60 120 Kilometers
88°36.00′E
88°48.00′E
89°0.00′E
89°12.00′E
89°24.00′E
89°36.00′E
24°24.00′N
24°12.00′N
24°0.00′N
23°48.00′N
Rajshahi
2425 km²
2.92 M
1804 km²
Pabna
2376 km²
2.91 M
1550 km²
Kushtia
1608 km²
2.15 M
1160 km²
India
Site 1
Site 2
Site 3
Site 5
Site 6
Site 7
Site 11
Site 12
Site 14
Site 15
Site 17
Site 26
Site 28
Site 31
Site 32
Site 34
Site 36
Site 39
Site 43
Site 44
Rooppur Nuclear Power Plant
Bheramara CC Power Station
0 10 20 30 km
Longitudinal Profile (Padma River-Sampling zone)
35 m
30 m
25 m
20 m
15 m
10 m
5 m
2 m
7.5 km
15 km
22.5 km
30 km
45 km
52.5 km
60 km
67.5 km
75 km
82.5 km
94.5 km

◂**Fig. 1** Study area overview depicting sampling locations, longitudinal river profile, and major infrastructure in the subtropical river basin, Padma River, Bangladesh. Main panel: 44 monitoring sites (orange circles) along a 94.5 km river stretch (yellow corridor) spanning Rajshahi, Pabna, and Kushtia districts (total area ~ 6600 km^2, population 8.3 million). The Rooppur Nuclear Power Plant and Bheramara CC Power Station are marked as key infrastructure. Bottom panel: Longitudinal elevation profile of the river course (2–35 m elevation range)

relevance to sensitive monitoring contexts such as the Padma River near the RNPP.

Materials and methods

Study area and hydrogeological setting

The Padma River Basin, located in western Bangladesh, plays a critical role in sustaining regional socioeconomic activities and agricultural livelihoods. The study area encompasses parts of the Kushtia, Rajshahi, and Pabna districts, covering approximately 6600 km^2 and supporting a population exceeding 8.3 million (Fig. 1). According to the Köppen climate classification, the basin experiences a humid subtropical climate characterized by hot summers, a pronounced monsoon season, and mild winters (Hossain et al., 2010).

Geologically, the basin forms part of the Ganges River floodplain and has been shaped by fluvial processes associated with the Ganges–Brahmaputra–Meghna River system (Usha et al., 2025). Surface deposits primarily consist of alluvial sands and estuarine sediments, underlain by a shallow and highly productive aquifer system. The terrain is largely flat, interspersed with shallow depressions and raised ridges that promote the formation of beels, canals, ponds, and seasonal wetlands. These landscape features function as natural water storage and recharge zones, maintaining surface–groundwater connectivity through overland flow and vertical infiltration.

The basin exhibits pronounced seasonal climatic variability. Air temperatures range from 25–36 °C during summer to 11–25 °C in winter (Uddin et al., 2021). Mean annual rainfall is approximately 1657 mm, with a peak in July (~ 336 mm) and a minimum in January (~ 8 mm). Evaporation rates vary between 17 and 140 mm, reflecting strong monsoonal influence. The groundwater table fluctuates seasonally between depths of 2 and 10 m, rising by approximately 3.5–5.5 m during the monsoon period (May–July) and declining gradually as recharge diminishes (Uddin et al., 2018). Groundwater flow generally follows a northwest-to-southeast direction across the basin (Uddin et al., 2017).

Groundwater serves as the primary source of water for domestic use (≈94%), as well as for agricultural and industrial activities. The Padma River, the principal surface water body in the basin, supports irrigation across nearly 4700 km^2 of cropland. As one of Bangladesh's most agriculturally intensive regions, the quality of both surface and groundwater resources is critical for food security, public health, and sustainable water management. Consequently, understanding the hydrogeochemical processes and spatiotemporal water quality dynamics of the Padma River is essential for developing effective monitoring strategies and evidence-based policy interventions in this monsoon-driven subtropical river system.

These physiographic and hydroclimatic characteristics directly informed the research design and methodological choices. The extreme seasonal contrast in rainfall (8 mm in January versus 336 mm in July) necessitated a two-season sampling strategy to capture maximum hydrochemical variability driven by monsoonal dilution and baseflow concentration processes. The high intensity of agricultural activity (4700 km^2 of irrigated cropland) and population density (8.3 million across 6600 km^2) justified the selection of nutrient and physicochemical parameters representing dominant anthropogenic stressors, including agricultural runoff and domestic wastewater discharge. Furthermore, the commissioning of the Rooppur Nuclear Power Plant (RNPP) in 2024—Bangladesh's first nuclear facility—created an urgent need for rapid baseline water quality assessment prior to full-scale operation. This motivated the establishment of 44 monitoring sites along a 94.5 km river corridor encompassing upstream reference areas, zones proximal to the RNPP, and downstream impact reaches. This spatial configuration enables the detection of localized contamination sources and the evaluation of longitudinal water quality gradients influenced by point-source discharges, diffuse agricultural inputs, and hydrological regime shifts between low-flow (winter) and high-flow (monsoon) conditions.

Sampling design, analytical methods, and quality assurance

A total of 88 surface water samples (44 samples per season: winter n = 44; monsoon n = 44) were collected from the Padma River Basin during the winter (January–February 2024) and monsoon (July–August 2024) seasons. Midstream samples (~0.3 m depth) were collected using pre-cleaned polyethylene bottles, preserved at 4 °C, and transported to the laboratory within 30 min of collection. The sampling strategy was designed to optimize spatial coverage while addressing practical constraints in establishing rapid baseline conditions near the newly operational Rooppur Nuclear Power Plant (RNPP), where multi-year monitoring was not logistically feasible during the 2024 commissioning period. In-situ parameters including temperature, electrical conductivity (EC), total dissolved solids (TDS), dissolved oxygen (DO), and salinity—were measured using a HANNA HI98194 multiparameter probe (Sowrav et al., 2024). The pH was determined separately with a Hanna HI-2211 pH meter for improved precision. All laboratory analyses followed APHA (2005) standard procedures. Major cations were quantified via Ion Chromatography (Dionex DX-3000, Thermo Fisher Scientific, USA) equipped with an IonPac CS17 analytical column, calibrated using the Dionex Six Cation-II standard. Anions and nutrients (NO_3^-, SO_4^{2-}, Cl^-, HCO_3^-, F^-) were analyzed with a UV–Vis spectrophotometer (HACH DR-3900) and titrimetric methods. Analytical quality was verified through routine calibration checks and replicate analysis (Alam et al., 2025). Total hardness, TSS, and TDS were determined through standard titrimetric and gravimetric methods.

Strict quality assurance and control (QA/QC) measures ensured analytical reliability. All reagents were of analytical grade (Fluka, Sigma-Aldrich, Germany), and ultrapure deionized water (18.2 MΩ cm, 0.2 μS cm^{-1}) was used throughout. Glassware was acid-washed (10% HNO_3) and rinsed thoroughly with deionized water. Instruments and pipettes were regularly calibrated under ISO/IEC 17025:2017 standards.

Non-acidified samples were used for anion analysis. Certified Reference Materials (CRMs) traceable to NIST (USA) verified analytical accuracy, while replicate analyses ensured precision. Additional standard solutions (Scharlau, Spain) were used to confirm instrument stability. Quality control included a sample blank, method blank, and CRM after every ten samples. Final concentrations represent the mean of triplicate determinations, with relative standard deviation (RSD) < 5% for all parameters.

Prior to modeling, all datasets were screened for completeness and consistency. Missing values accounted for less than 3% of the total observations and were handled using season-specific median imputation to preserve distributional characteristics while avoiding artificial variance inflation. Outliers were identified using interquartile range (IQR) and z-score thresholds ($|z| > 3$) and were retained unless physically implausible, ensuring that extreme but realistic hydrological events were not excluded. Measurement uncertainty arising from instrumental precision, laboratory analysis, and sampling variability was implicitly propagated through the modeling framework via cross-validation, Physics-Informed Neural Network (PINN)-based data augmentation, and Monte Carlo simulation, thereby minimizing bias in predictive inference.

Data completeness was systematically verified using a custom Python-based quality assurance script (DataFrame_Checker) which confirmed zero missing values, zero duplicate samples, and 100% data completeness across all 18 physicochemical parameters for both winter and monsoon seasons (n = 88 total samples; 44 per season). All 1584 individual measurements (18 parameters × 88 samples) were successfully quantified above instrument detection limits, with no censored or below-detection-limit values requiring imputation or substitution. This complete dataset reflects rigorous field sampling protocols (midstream collection at 0.3 m depth, immediate preservation at 4 °C, transport to laboratory within 30 min). Table S12 provides instrument methods, temporal coverage, spatial density, and data completeness metrics for all measured parameters, ensuring full transparency and reproducibility of the water quality assessment.

Water quality index (WQI) calculation

The Water Quality Index (WQI) provides an integrated assessment of surface water suitability for various uses and is widely employed in environmental monitoring and management. Among several established models such as NSF-WQI, CCME-WQI, and

Oregon-WQI, this study adopted the Weighted Arithmetic Water Quality Index (WA-WQI) method (Horton, 1965) for its simplicity, robustness, and applicability across diverse hydrological contexts (Islam et al., 2025).

The WA-WQI aggregates multiple physicochemical parameters into a single index value can be found in Table S2.

Standard limits and ideal values were adopted from the Bangladesh National Drinking Water Standards and relevant international guidelines (DoE, 2010; Ahsan et al., 2023). Parameter settings and standard values are summarized in Table S3, with classification criteria detailed in Table S2.

WQI was selected over alternative indices such as the Canadian Council of Ministers of the Environment Water Quality Index or the Oregon Water Quality Index because it provides site-specific sensitivity to local water quality standards through the use of WHO (2017) guideline values as reference benchmarks. Unlike simple averaging approaches, the weighted arithmetic mean accounts for differential parameter importance while remaining computationally transparent, which is critical for stakeholder communication and regulatory compliance in Bangladesh's water management framework.

Advanced predictive modeling framework

Architecture overview

The predictive modeling framework (Fig. 2) comprised five sequential phases:

1. Data preprocessing
2. Feature engineering
3. Data augmentation
4. Model training and evaluation
5. Interpretability and generalization assessment

Data preprocessing involved exploratory analysis, normalization, and identification of anomalies

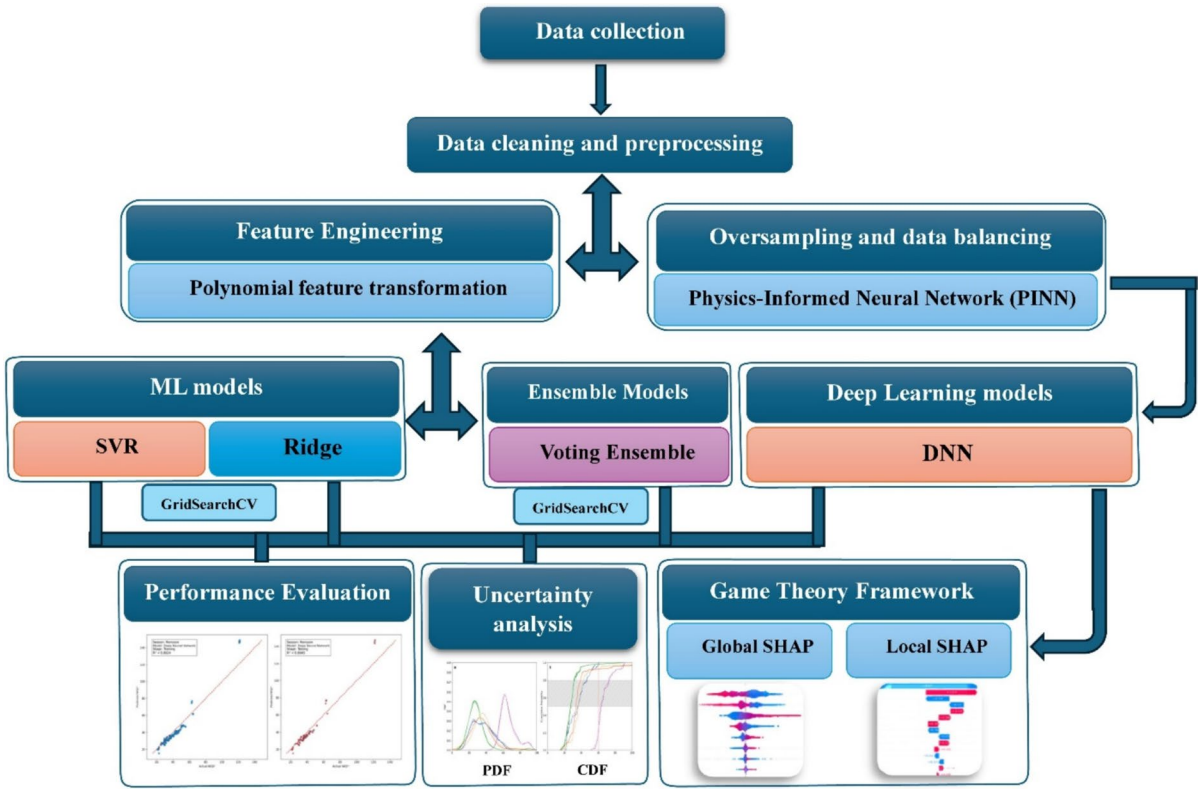


Fig. 2 Proposed integrated AI-based modelling workflow for seasonal WQI prediction adopted for this research

and seasonality. To capture nonlinear interactions, Polynomial Feature Transformation (degree = 2) expanded 18 input variables into 189 features, including squared and pairwise terms (Tanoli et al., 2024).

The machine learning phase incorporated Support Vector Regression (SVR), Ridge Regression, and a Voting Ensemble, each optimized via cross-validated GridSearchCV (Table S4). Seasonal datasets were further augmented using a Physics-Informed Neural Network (PINN) to address limited samples while maintaining hydrochemical consistency. Supplementary Text S1 shows the justification for selecting 18 features in ML and ensemble models and 20 features in DNN and PINN models. Data were split 80:20 for training and testing, following established practice (Khosravi et al., 2025; Mamun et al., 2024). The deep learning models were trained on augmented datasets (440 total samples per season).

Machine learning models

Support vector regression (SVR) SVR, introduced by Vapnik (1995), was used to model nonlinear relationships in WQI. Despite higher computational costs, SVR performs reliably on small-to-medium datasets and is resistant to overfitting. The model employed a Radial Basis Function (RBF) kernel and polynomial feature expansion to enhance flexibility. Hyperparameters were optimized using GridSearchCV (Imam et al., 2024; Mamun et al., 2024). Table S5 reveals configuration of features and target variables in machine learning applied for this research. SVR was chosen for its capacity to describe nonlinear connections via kernel changes, while maintaining robustness to outliers via the ε-insensitive loss function. An RBF kernel was determined to be optimum by grid-search optimization (Table S4). SVR does not need assumptions about error distributions, making it particularly appropriate for environmental data characterized by episodic extremes, such as TSS surges induced by monsoons. Seasonal stationarity was managed by developing distinct models for winter and monsoon, while feature dependency was analyzed by principal component analysis (PCA); hitherto, polynomial features were maintained to ensure interpretability despite existing connections.

Ridge regression Ridge Regression (L2-regularized) effectively addresses multicollinearity and noise sensitivity, producing stable estimates for correlated predictors (Proshad et al., 2025). Polynomial expansion was applied to introduce nonlinearity, and parameter tuning was performed through cross-validation (Shams et al., 2024). Ridge Regression was chosen for its effectiveness in handling multicollinearity among the 189 polynomial features through L2 regularization, which shrinks correlated coefficients without eliminating features. This capability is critically given the strong correlations among hydrochemical parameters (e.g., EC–TDS, $r \approx 1.00$; Fig. S7). Ridge assumes linear relationships between predictors and WQI, which is reasonable for physicochemical parameters within WHO guideline ranges, although it may underrepresent nonlinear interactions. The model's key assumptions, homoscedasticity, independence of residuals, and approximate normality were assessed using residual diagnostics, which confirmed uniform error distribution across predicted WQI values.

Ensemble learning

A Weighted Soft Voting Ensemble was developed to combine the strengths of SVR (nonlinear adaptability) and Ridge Regression (linear stability) (Satish et al., 2024). Model weights were assigned based on seasonal performance to balance bias and variance. Performance evaluation employed fivefold cross-validation, and interpretability was enhanced through SHAP analysis. The Voting Ensemble, combining Ridge Regression, SVR, and Random Forest, was employed to integrate linear efficiency, nonlinear modeling capacity, and robustness from bootstrap aggregation. Ensemble averaging reduces individual model biases and improves prediction stability across varying hydrological conditions. Model diversity, a key ensemble requirement, was verified by low inter-model prediction correlations ($r < 0.95$ for all model pairs), representing sufficient complementarity for effective ensemble learning.

Physics-Informed Neural Network (PINN)

PINNs, introduced by Raissi et al. (2017), integrate physical laws into neural network training. Here, PINNs were used for data augmentation, generating realistic synthetic samples while preserving hydrochemical relationships. The models were implemented in PyTorch using an encoder–latent–decoder

architecture, trained for 1000 epochs with batch size = 8 and Adam optimizer. A schematic of the PINN architecture and loss function design is shown in Fig. S1.

To address the n = 44 limitation for deep learning model training, Physics-Informed Neural Networks (PINNs) generated 10 × synthetic samples per season (440 samples per season), preserving statistical moments (mean, standard deviation, skewness, kurtosis; see Tables S6–S7 and Figs. S2–S3 in Supplementary Materials). For each season, 10× synthetic samples were produced, expanding the dataset while maintaining statistical integrity. Comparisons of mean, SD, skewness, and kurtosis between real and synthetic data (Tables S6 and S7; Figs. S2 and S3) confirmed dataset reliability and representativeness.

Deep neural network (DNN)

DNN models were constructed in TensorFlow 2.18.0 (Keras backend) to predict seasonal WQI (WQI^w and WQI^m). The architecture included an input layer, three hidden layers (128, 64, 32 neurons), and an output neuron. Hidden layers used ReLU activation, L1–L2 regularization (λ = 0.001), batch normalization, and dropout (rate = 0.3) for stability. Models were trained for up to 200 epochs (batch size = 16) with Adam optimizer (learning rate = 0.001), early stopping, and checkpointing to prevent overfitting (Gao et al., 2025; Hu et al., 2023) (Fig. S4). Configuration of features and target variables in deep learning models can be found elsewhere (Table S5). DNN were selected for their capacity to learn hierarchical nonlinear feature representations without explicit feature engineering, which is particularly important for complex ion interaction effects such as pH-NO_3^--SO_4^{2-} interactions affecting WQI. The 128-64-32 architecture with dropout regularization (rate = 0.3) and combined L1–L2 regularization (λ = 0.001) was optimized via systematic grid search (Table S4). DNN training assumes sufficient training data (addressed via PINN augmentation (Raissi et al., 2017) generating 440 samples per season), stationarity of learned representations within seasons (satisfied by separate seasonal models), and independence of training samples (ensured through random train-test splitting with 80–20 ratio).

Game theory-based interpretability

To quantify variable influence on DNN predictions, a Game Theory-based framework was integrated. Rooted in Shapley value theory (von Neumann & Morgenstern, 1944), this method evaluates each feature's marginal contribution to WQI prediction. A custom Game Theory WQI Analyzer (Algorithm 1, supplementary Text 2) was implemented using Monte Carlo simulations of parameter permutations. Shapley values were averaged across test samples to obtain robust feature importance rankings, complementing SHAP-based interpretability (Talukdar et al., 2024).

SHAP-based model

SHAP (SHapley Additive Explanations) was used to interpret DNN predictions and visualize feature importance. Two interpretive visualizations were generated: Beeswarm plots show global feature importance and variability. Waterfall plots illustrate individual feature contributions for specific predictions (Mamun et al., 2024). Together with the game-theory approach, SHAP analysis ensured model transparency and explainability, providing interpretable insights for decision-makers.

Model Performance Evaluation

Model performance was assessed using multiple statistical metrics: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R^2), and Relative Absolute Error (RAE). These metrics collectively evaluated accuracy, reliability, and generalization performance across the SVR, Ridge, Voting Ensemble, and DNN models (Islam et al., 2025; Nguyen et al., 2024).

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\mathrm{WQI}_a - \mathrm{WQI}_p\right)^2} \tag{1}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(\mathrm{WQI}_a - \overline{\mathrm{WQI}}_p\right)^2}{\sum_{i=1}^{n}\left(\mathrm{WQI}_a - \overline{\mathrm{WQI}}_a\right)^2} \tag{2}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|\mathrm{WQI}_a - \mathrm{WQI}_p\right| \tag{3}$$

$$\mathrm{RAE} = \frac{\sum_{i=1}^{n} \left|\mathrm{WQI}_a - \mathrm{WQI}_p\right|}{\sum_{i=1}^{n} |\mathrm{WQI}_a - \overline{\mathrm{WQI}_a}|} \tag{4}$$

$$\mathrm{NSE} = 1 - \frac{\sum_{i=1}^{n} \left(\mathrm{WQI}_a - \mathrm{WQI}_p\right)^2}{\sum_{i=1}^{n} \left(\mathrm{WQI}_a - \overline{\mathrm{WQI}_a}\right)^2} \tag{5}$$

$$\mathrm{MEF} = 1 - \frac{\sum_{i=1}^{n} \left|\mathrm{WQI}_a - \mathrm{WQI}_p\right|}{\sum_{i=1}^{n} |\,\mathrm{WQI}_a - \overline{\mathrm{WQI}_a}\,|} \tag{6}$$

WQI_a and WQI_p are the actual and predicted WQI values, respectively. Furthermore, $\overline{\mathrm{WQI}_a}$ and $\overline{\mathrm{WQI}_p}$ are the average values of the actual and predicted values, respectively. The variable n denotes total number of observations.

Model efficiency and uncertainty analysis

Model performance, robustness, and uncertainty were comprehensively evaluated to validate the predictive accuracy of WQI estimates across the Padma River basin. Model efficiency was assessed using the Nash–Sutcliffe Efficiency (NSE) and Model Efficiency Factor (MEF), where higher values indicate stronger predictive agreement and robustness (Mamun et al., 2024). Both metrics were computed for winter and monsoon seasons to assess temporal generalization.

Prediction uncertainty was analyzed using probability density functions (PDFs) and cumulative distribution functions (CDFs) derived from observed and predicted WQI datasets. Kernel Density Estimation (KDE) was employed to generate smooth probability distributions, facilitating direct comparison between predicted and observed data (Gelman et al., 2022). These analyses characterized model bias, spread, and seasonal confidence intervals.

Model robustness was further validated through fivefold cross-validation (Radočaj & Jurišić, 2025). The mean and standard deviation of the coefficient of determination (R^2) across folds were calculated as.

$$\mathrm{CV}_{R^2}^{\mathrm{mean}} = \frac{1}{k} \sum_{j=1}^{k} R_j^2 \tag{7}$$

$$\mathrm{CV}_{R^2}^{\mathrm{SD}} = \sqrt{\frac{1}{k-1} \sum_{j=1}^{k} \left(R_j^2 - \mathrm{CV}_{R^2}^{\mathrm{mean}}\right)^2} \tag{8}$$

where R_j^2 represents coefficient of determination for the jth fold, k denotes number of cross-validation folds (here, $k = 5$), $\mathrm{CV}_{R^2}^{\mathrm{mean}}$ represents average R^2 across folds and $\mathrm{CV}_{R^2}^{\mathrm{SD}}$ reflects standard deviation of R^2 across folds. These metrics (Eqs. 7 and 8) provide a measure of the consistency and variance of model performance across different subsets of the training data.

Monte Carlo Simulation (MCS)

Baseline Water Quality Index (WQI) values were first estimated using the developed predictive models. Monte Carlo Simulation (MCS) was then applied to quantify uncertainty and characterize the probabilistic behavior of WQI during winter and monsoon seasons. A lognormal distribution, suitable for positively skewed environmental data (Islam et al., 2025), was fitted using seasonal mean (μ) and standard deviation (σ) derived from observed WQI values. Each simulation generated 20,000 random iterations to ensure statistical robustness and capture variability. The resulting probability density (PDF) and cumulative distribution (CDF) functions depicted key percentiles (25th–95th), providing a comprehensive visualization of uncertainty and seasonal differences in water quality dynamics across the Padma River basin.

Spatial regionalization and spatial autocorrelation analysis

A spatial network connectivity framework was developed to quantify spatial regionalization relationships among similarly classified WQI zones. Using Euclidean distance-based connectivity, a binary adjacency matrix was generated to represent spatial interactions between monitoring sites (Proshad et al., 2024).

The connectivity weight between any two sampling sites i and j was defined as:

$$W_{ij} = \begin{cases} 1, & \text{if } d(i,j) \le d_{\max} \text{ and } C_i = C_j \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

where:

- $d(i,j)$ = Euclidean distance between sites i and j,
- d_{max} = maximum threshold distance for spatial connection, and
- Ci and Cj = WQI classification categories for sites i and j, respectively.

A seasonal transition matrix was then used to track WQI class shifts between winter and monsoon, revealing spatial coherence and temporal variability across the basin.

Spatial dependence was quantified using Moran's I and Local Indicators of Spatial Association (LISA) to detect high–high (HH), low–low (LL), and outlier (HL, LH) clusters (Proshad et al., 2025). Inverse Distance Weighted (IDW) interpolation in ArcGIS 10.8 visualized the spatial distribution of WQI and key parameters, highlighting critical zones near the Ruppur Nuclear Power Plant. Additionally, a Sankey diagram illustrated seasonal transitions between WQI classes, emphasizing the resilience, vulnerability, and recovery dynamics of distinct river segments.

Local Indicators of Spatial Association (LISA) and Moran's I were selected over alternative spatial statistics such as Getis-Ord Gi* or semivariogram analysis because LISA provides dual functionality by identifying both clustering patterns and spatial outliers (high-low and low–high anomalies), which offers richer information for pollution source identification than methods focused solely on hotspot detection. Global Moran's I complement local analysis by quantifying overall spatial autocorrelation across the entire basin, validating whether local clustering patterns are statistically significant at the basin scale rather than representing random spatial noise. Spatial pattern robustness was tested using multiple distance thresholds (5, 10, 15 km) to verify that observed clustering was not an artifact of arbitrary neighborhood definition. Appropriate neighborhood definition requires careful selection of the spatial distance threshold for defining "neighboring" sites. This assumption was verified through sensitivity analysis using distance thresholds of 5 km, 10 km, and 15 km (Table S14). Statistical significance of spatial autocorrelation was assessed using 999 Monte Carlo permutations rather than analytical p-values, which accounts for spatial autocorrelation in the null hypothesis generation and provides more conservative significance testing appropriate for spatially structured environmental data.

Statistical data analysis and verification

Descriptive statistics (mean, SD, CV, quartiles, and skewness) were used to characterize seasonal variations in physicochemical parameters of the Padma River basin (Uddin et al., 2024) (Tables S8 and S9). Frequency distributions and summary statistics illustrated spatial and temporal variability between winter and monsoon seasons (Figs. S5 and S6). One-way ANOVA tested seasonal and spatial differences, with significance defined at $p < 0.05$ ($F_{(calculated)} > F_{(critical)}$) (Ravi et al., 2023). Pearson's correlation quantified linear relationships among parameters and the Water Quality Index (WQI), where $r > 0.7$ indicated strong positive associations (Figs. S7 and S8). Hierarchical clustering (Ward's method) and correlation heatmaps visualized inter-parameter linkages and seasonal groupings. Z-score normalization enabled cross-seasonal comparability, and boxplots depicted standardized variability. Statistical Process Control (SPC) tools (e.g., X-bar, EWMA, CUSUM, and control charts) monitored data stability and identified anomalies (Fig. S9). All analyses were conducted in Python 3.12.7 (Jupyter Notebook v7.2.2) using pandas, numpy, matplotlib, seaborn, scikit-learn, TensorFlow, and SHAP. Spatial analyses (IDW, LISA, Moran's I) were performed in ArcGIS 10.8.

The model and statistical assumptions were rigorously assessed to guarantee analytical validity. ANOVA was chosen for seasonal comparisons because of equal sample sizes (n = 44 per season) and substantial effect sizes, evidenced by exceptionally high F-statistics ($F > 100$, $p < 0.0001$), which demonstrate that seasonal variability significantly surpasses within-season variance and maintains robustness against moderate assumption deviations. Independence was guaranteed by geographic randomization of sample locations throughout the 94.5 km river corridor and a separation of almost five months between winter (January–February) and monsoon (July–August) sampling operations. The homogeneity of variance was sufficient for most parameters, and small breaches were addressed by equal sample sizes and extreme F-values. Non-normally distributed variables underwent log transformation prior to analysis.

Regression assumptions were validated by residual diagnostics, which indicated consistent error distributions throughout anticipated WQI ranges, thus validating homoscedasticity. Sensitivity analysis at

distance limits of 5, 10, and 15 km revealed substantial winter clustering ($p < 0.001$) and scale-dependent monsoon clustering at 15 km ($p = 0.027$; Table S14). Seasonal non-stationarity was mitigated by developing distinct models for winter and monsoon circumstances, whereas within-season stationarity was deemed acceptable due to short sample periods (6–8 weeks) lacking significant hydrological disruptions. The verification procedures, comprising residual analysis, spatial statistics, cross-validation stability, and robust ANOVA signals, collectively affirm the reliability of the analytical results and suggest that the observed patterns represent authentic hydrochemical variability rather than methodological artifacts.

Results

Basic characteristics and seasonal variation of water quality indicators

The physicochemical composition of surface water across the Padma River exhibits pronounced seasonal contrasts (Table 1; Figs. S5–S6). The pH varied from 5.7 to 7.8 (mean: 6.6 ± 0.4) in winter—slightly acidic to neutral to 6.4–8.1 (7.7 ± 0.3) in the monsoon, trending toward neutral to mildly alkaline, remaining within WHO's acceptable range. In winter, parameters such as electrical conductivity (EC), total dissolved solids (TDS), total hardness (TH), and major ions reach higher levels, reflecting reduced river discharge, limited dilution, and evaporative enrichment typical of low-flow conditions. Conversely, the monsoon season shows significant reductions in these solutes as a result of hydrological flushing and increased catchment runoff, consistent with a dilution-dominated regime.

Dissolved oxygen (DO) decreases markedly during the monsoon (Table 1; Figs. S5–S6), likely due to increased organic matter loads and oxygen demand from microbial decomposition under high-flow, turbid conditions. Total suspended solids (TSS) increase substantially, indicating enhanced sediment resuspension and soil erosion from surrounding landscapes during heavy rainfall. These patterns collectively indicate that monsoon forcing alters water quality through both dilution and mobilization of particulate and organic loads, rather than dilution alone.

Nutrient concentrations, particularly nitrate, exhibit clear seasonal variability. While winter shows relatively stable NO_3^- levels, the monsoon season displays broader variability and episodic elevation, suggesting that agricultural runoff and diffuse source inputs interact with streamflow dynamics to influence

Table 1 Physicochemical parameter concentrations during winter and monsoon seasons in the Padma River basin: statistics including range, mean ± SD, and WHO (2017) standards

Parameters		Winter		Monsoon	
	WHO (2017)	Range	Mean ± SD	Range	Mean ± SD
Temperature (°C)	–	22.0–32.0	25.6 ± 1.9	26.0–32.0	30.2 ± 1.8
pH	6.5–8.5	5.7–7.8	6.6 ± 0.4	6.4–8.1	7.7 ± 0.3
EC (μS/cm)	1500	240.0–641.0	379.8 ± 71.5	114.0–357.0	212.3 ± 32.9
TDS (mg/dm^3)	1000	143.7–384.6	227.9 ± 42.9	63.2–214.2	116.9 ± 20.0
DO (mg/dm^3)	5	14.1–19.4	17.4 ± 1.3	0.0–18.1	0.8 ± 2.7
Salinity (%)	–	0.1–0.4	0.2 ± 0.1	0.2–4.1	3.8 ± 0.6
TSS (mg/dm^3)	30	1.4–379.0	81.0 ± 64.9	0.5–2319.6	593.0 ± 430.7
TH (mg/dm^3)	100	227.2–709.8	356.0 ± 96.8	186.0–630.0	264.2 ± 85.3
HCO_3^- (mg/dm^3)	300	128.0–356.0	170.9 ± 46.7	90.0–158.0	122.9 ± 13.5
F^- (mg/L)	1.5	0.0–0.4	0.2 ± 0.1	0.0–0.0	0.0 ± 0.0
Cl^- (mg/L)	250	3.6–14.1	11.8 ± 3.3	2.3–21.7	3.0 ± 2.9
NO_3^- (mg/L)	50	0.5–130.5	24.0 ± 36.3	3.6–454.4	47.5 ± 68.7
SO_4^{2-} (mg/L)	250	3.8–17.7	15.2 ± 3.9	6.0–34.2	7.2 ± 4.2
Na^+ (mg/L)	200	11.7–58.6	22.4 ± 6.7	6.0–48.9	7.9 ± 6.3
K^+ (mg/L)	30	1.3–8.5	4.5 ± 1.0	1.0–3.6	3.1 ± 0.4
Mg^{2+} (mg/L)	150	12.1–29.2	15.0 ± 4.0	5.0–24.7	5.7 ± 2.9
Ca^{2+} (mg/L)	200	16.1–60.1	34.6 ± 8.2	25.7–139.2	29.8 ± 16.9

nutrient distribution. Other major ions including sulfate (SO_4^{2-}), sodium (Na^+), potassium (K^+), magnesium (Mg^{2+}), and calcium (Ca^{2+}) largely remain within WHO guideline limits but display higher winter concentrations consistent with reduced flushing and evaporative concentration.

Seasonal variability based on standardized Z-scores

Standardized Z-score profiles (Fig. 3) reinforce that seasonal shifts are systematic and parameter-specific rather than random. Winter dominance of ionic constituents and TH aligns with solute accumulation processes under low-flow conditions, whereas monsoon amplification of TSS and temperature reflects runoff-driven sediment influx and thermal forcing. Importantly, the Z-scores for the Water Quality Index (WQI) indicate lower overall water quality during winter relative to the monsoon (Fig. 3s), directly supporting the hypothesis that seasonal hydrology is the primary driver of integrated water quality outcomes.

Seasonal and spatial effects using ANOVA test

The analysis of variance (Table 2) demonstrates that temporal (seasonal) variability overwhelmingly outweighs spatial variability in controlling water quality. Most parameters show highly significant seasonal differences, confirming strong monsoon forcing. Calcium exhibits limited seasonal sensitivity, indicating dominant geogenic control, whereas nitrate shows marginal seasonal significance, consistent with counteracting effects of runoff enrichment and dilution. Spatial variability is generally weak, with significant effects confined mainly to chloride and sulfate (Table 2). This pattern points to localized anthropogenic or geochemical inputs rather than basin-wide spatial structuring, reinforcing the conclusion that seasonal hydroclimatic forcing dominates over site-specific effects in shaping water quality.

NO_3^- exhibited marginal seasonal significance ($F = 4.02$, $p = 0.048$), whereas Ca^{2+} demonstrated no statistically significant seasonal variation ($F = 2.88$, $p = 0.094$), indicating its relative stability across hydrological regimes.

In contrast, spatial variations across sampling sites were generally weaker. Only pH ($F = 4.36$, $p = 0.024$), chloride (Cl^-, $F = 26.64$, $p < 0.0001$), and sulfate (SO_4^{2-}, $F = 3.85$, $p = 0.035$) exceeded the $F_{(critical)}$ (3.35) threshold, indicating localized influences such as geochemical heterogeneity or point-source discharges. Other parameters, such as DO and TSS, came close to being statistically significant but didn't quite make it. This shows that the water quality in the basin was pretty evenly spread out in winter and monsoon season (Figs. S10 and S11).

Within the 2024 sampling period, ANOVA results indicate statistically detectable differences between winter and monsoon conditions for most parameters ($p < 0.05$), whereas spatial contrasts across sampling locations were comparatively weaker. These results reflect within-year seasonal contrasts under the observed hydrological conditions rather than persistent dominance of seasonality over spatial variability.

Post-hoc effect size analysis confirmed substantial seasonal influence (median $\eta^2 = 0.61$; range 0.03–0.95) with adequate statistical power ($1\text{-}\beta > 0.95$ for 88% of parameters; Table S13). Cohen's f values ranged from 0.18 (Ca^{2+}) to 4.28 (Salinity), with dissolved oxygen ($f = 3.97$) and salinity ($f = 4.28$) exhibiting very large effect sizes, validating the robustness of seasonal differentiation observed in the ANOVA results.

Performance assessment and validation of XAI-enhanced ML and DL models

Model performance results are presented in Table 3 and Fig. 4. Clear hydrological regime dependence is evident. During winter, the deep neural network (DNN) achieves superior predictive accuracy, reflecting its ability to capture nonlinear relationships under relatively stable low-flow conditions. In contrast, its reduced performance during the monsoon indicates sensitivity to increased nonstationarity and hydrological variability. Ensemble and regularized models (Voting Ensemble and Ridge Regression) demonstrate greater robustness during the monsoon (Table 3; Fig. 4), suggesting that model averaging and regularization better accommodate dynamic flow regimes. These results directly address Objective (i), demonstrating that no single modeling approach is universally optimal and highlighting the need for season-adaptive predictive frameworks in monsoon-driven river systems..

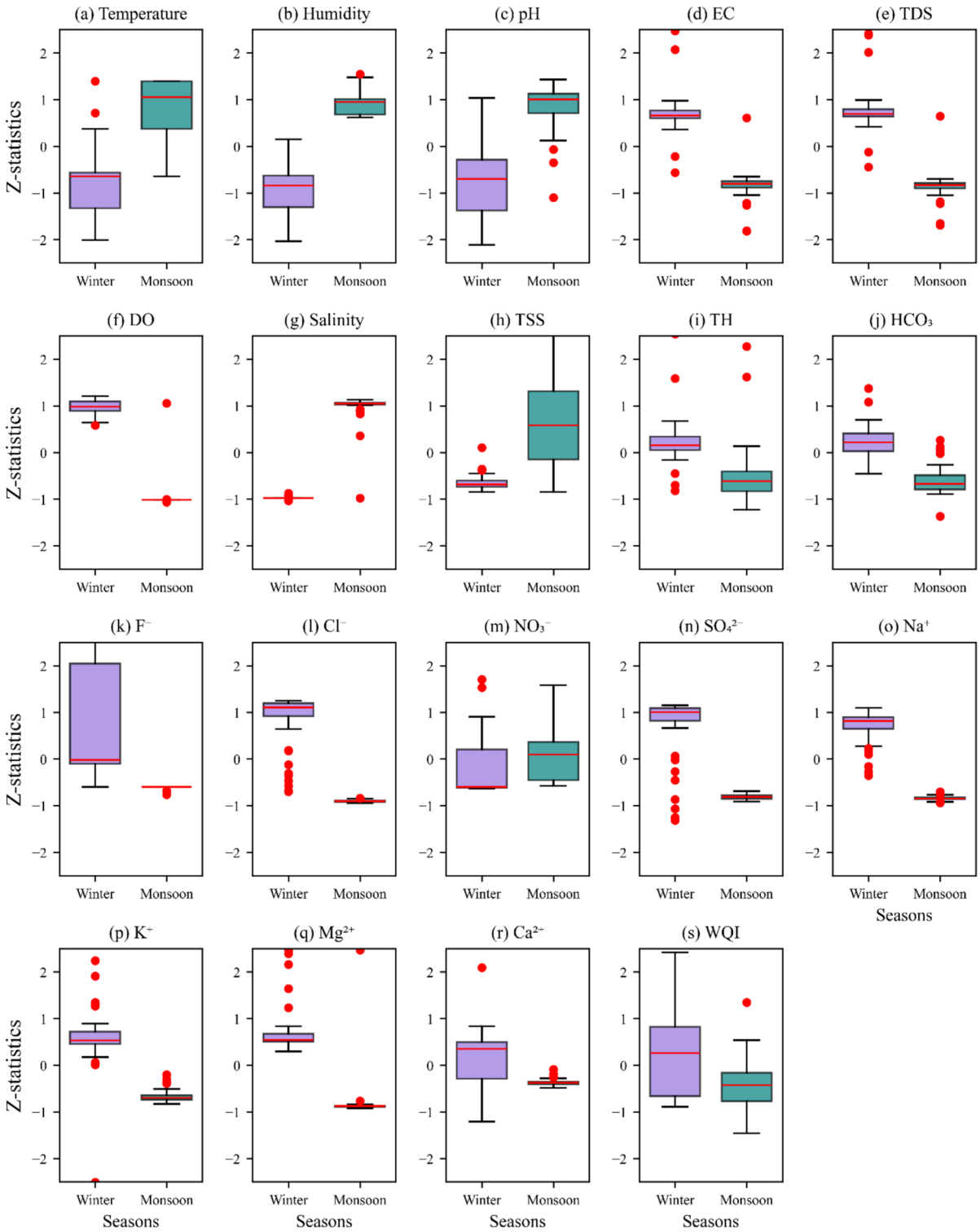


Fig. 3 Standardized z-statistics of seasonal water quality indicators and WQI in the study river basin. Purple and teal boxplots represent winter and monsoon seasons, respectively, with red dots indicating outliers beyond 1.5× IQR. Higher Z-scores reflect greater deviation from the mean, showing seasonal concentration effects (winter) or dilution/enhanced inputs (monsoon)

Table 2 Results of analysis of variance (ANOVA) for seasonal and spatial variations in physicochemical parameters of the Padma River basin

Parameter	$F_{calculated}$	*p*-value	$F_{critical}$
Seasonal variation			
Temperature	132.06	< 0.0001	3.95
pH	176.45	< 0.0001	3.95
EC	199.22	< 0.0001	3.95
TDS	242	< 0.0001	3.95
DO	1358.64	< 0.0001	3.95
Salinity	1573.43	< 0.0001	3.95
TSS	60.8	< 0.0001	3.95
TH	22.29	< 0.0001	3.95
HCO_3^-	42.87	< 0.0001	3.95
F^-	49.29	< 0.0001	3.95
Cl^-	178.76	< 0.0001	3.95
NO_3^-	4.02	0.048	3.95
SO_4^{2-}	86.43	< 0.0001	3.95
Na^+	109	< 0.0001	3.95
K^+	84.82	< 0.0001	3.95
Mg^{2+}	155.95	< 0.0001	3.95
Ca^{2+}	2.88	0.094	3.95
Spatial variation			
Temperature	1.45	0.321	3.35
pH	4.36	0.024	3.35
EC	0.46	0.941	3.35
TDS	0.46	0.941	3.35
DO	3.35	0.050	3.35
Salinity	0.52	0.908	3.35
TSS	3.07 ara>	0.063	3.35
TH	0.56	0.883	3.35
HCO_3^-	0.48	0.931	3.35
F^-	2.04	0.166	3.35
Cl^-	26.64	< 0.0001	3.35
NO_3^-	1.8	0.215	3.35
SO_4^{2-}	3.85	0.035	3.35
Na^+	0.34	0.985	3.35
K^+	0.95	0.584	3.35
Mg^{2+}	0.33	0.987	3.35
Ca^{2+}	2.04	0.164	3.35

Model performance for machine learning (ML), deep learning (DL), and ensemble approaches was evaluated separately for winter and monsoon seasons using six metrics: R^2, RMSE, MAE, RAE, NSE, and MEF (Table 3).

During winter, the DNN model achieved the best predictive accuracy ($R^2 = 0.9837$, RMSE = 1.16), indicating excellent generalization and minimal error. In contrast, traditional models—Ridge Regression ($R^2 = 0.7897$, RMSE = 6.08) and Support Vector Regression (SVR) ($R^2 = 0.8591$, RMSE = 4.97)—showed moderate performance with signs of overfitting. The Voting Ensemble exhibited intermediate accuracy ($R^2 = 0.8519$, RMSE = 5.10) (Fig. 4).

In monsoon, performance patterns reversed. The Voting Ensemble produced the most robust predictions ($R^2 = 0.9649$, RMSE = 1.07), followed closely by Ridge Regression ($R^2 = 0.9681$, RMSE = 1.02) and SVR ($R^2 = 0.9526$, RMSE = 1.24). The DNN exhibited reduced adaptability ($R^2 = 0.8840$, RMSE = 6.02), likely due to higher hydrological variability and increased nonstationarity during monsoon flows (Fig. 4).

Overall, these findings emphasize the context-dependent performance of predictive models. The DNN excelled under stable winter conditions, effectively capturing nonlinear relationships, while ensemble methods proved more resilient to the dynamic, uncertain hydrological conditions of the monsoon season. This illustrates the importance of adaptive modeling frameworks that align algorithmic strengths with seasonal variability in river systems.

All models were validated using fivefold cross-validation (Table S10). To enhance transparency and interpretability, SHAP values were computed for both global and local feature importance, and a game theory-based Shapley framework quantified each parameter's relative contribution to WQI prediction.

Learning curve analysis (Fig. S12) confirms successful model convergence without overfitting for both seasonal DNNs. Winter season model exhibits smooth convergence with training and validation loss plateauing after approximately 50 epochs. Monsoon season model converges earlier (approximately 40 epochs) but displays greater post-convergence fluctuation in validation metrics, consistent with higher measurement variance in monsoon conditions (dissolved oxygen CV = 343% vs. 7.7% in winter; Tables S8–S9). Critically, validation curves in both seasons converge with training curves without systematic divergence, demonstrating that dropout regularization (rate = 0.3), L1–L2 regularization ($\lambda = 0.001$), and early stopping successfully prevented overfitting despite the 128–64-32 architecture operating on 440

Table 3 Performance evaluation of ML, DL, and ensemble models for river water quality prediction across winter and monsoon seasons

	Ridge		SVR		Voting ensemble		DNN	
Metric	Train	Test	Train	Test	Train	Test	Train	Test
Winter season								
R^2	0.9975	0.7897	0.9937	0.8591	0.9952	0.8519	0.9809	0.9837
RMSE	0.5319	6.0754	0.8435	4.9730	0.7384	5.0982	1.1704	1.1557
RAE	0.0419	0.2180	0.0315	0.2612	0.0338	0.2473	0.9251	0.9020
MAE	0.3707	2.4640	0.2787	2.9516	0.2995	2.7947	0.1417	0.1294
NSE	0.9975	0.7897	0.9937	0.8591	0.9952	0.8519	0.9809	0.9837
MEF	0.9581	0.7820	0.9685	0.7388	0.9662	0.7527	0.8582	0.8705
Monsoon season								
R^2	0.9944	0.9681	0.9991	0.9526	0.9987	0.9649	0.8624	0.8840
RMSE	0.9284	1.0175	0.3706	1.2407	0.4472	1.0668	5.5566	6.0217
RAE	0.1031	0.1832	0.0181	0.2449	0.0388	0.2057	3.9716	4.1011
MAE	0.7084	0.8320	0.1244 ara>	1.1122	0.2666	0.9342	0.4304	0.4270
NSE	0.9944	0.9681	0.9991	0.9526	0.9987	0.9649	0.8624	0.8840
MEF	0.8969	0.8168	0.9819	0.7551	0.9612	0.7943	0.5696	0.5730

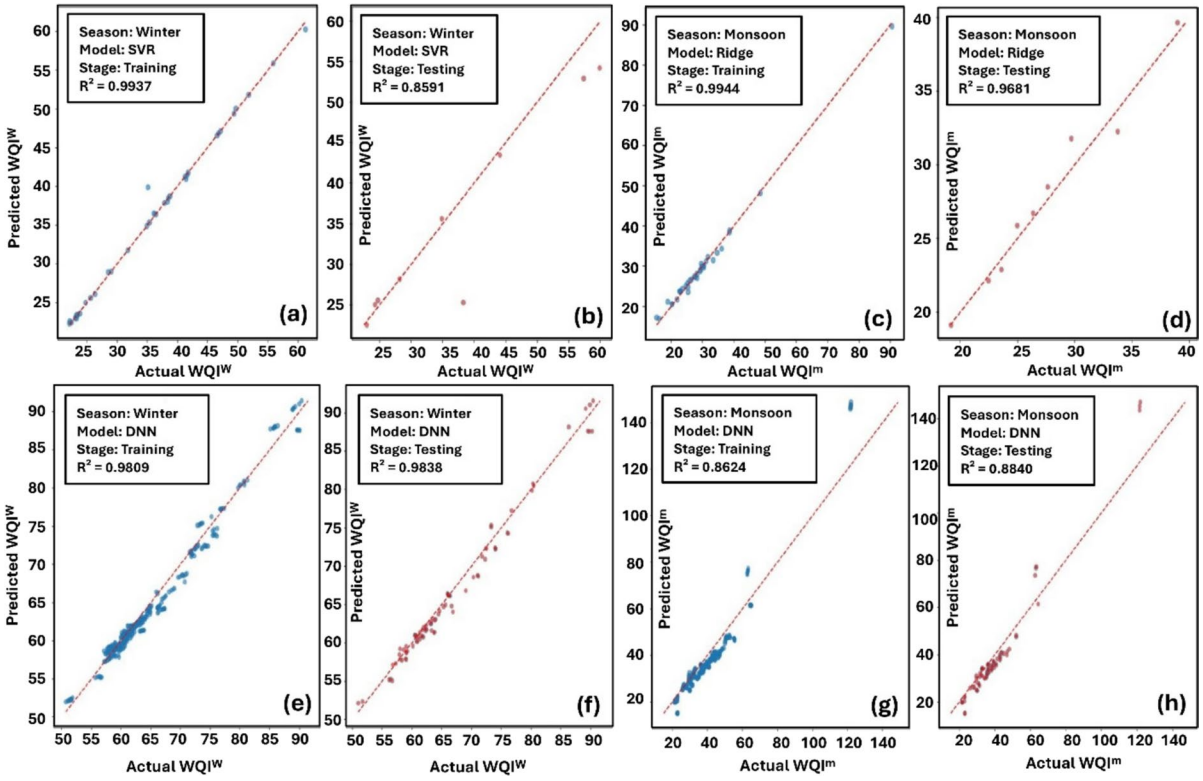


Fig. 4 Actual versus predicted WQI values for the best-performing machine learning (SVR, Ridge regression) and deep learning (DNN) models during winter and monsoon seasons in the Padma River basin. Panels (a–b) show SVR performance for winter (training and testing, respectively), panels (c–d) show Ridge regression performance for monsoon (training and testing, respectively), and panels (e–h) show DNN performance for winter and monsoon under training and testing stages. The red dashed line represents the 1:1 line (perfect agreement), and model/season/stage information with R^2 is provided within each panel

PINN-augmented samples per season. Table 3 reflects genuine hydrological complexity rather than model overfitting, as evidenced by the absence of training-validation divergence in learning curves.

SHAP analysis using the Game theory model

Although the DNN (winter) and Voting Ensemble (monsoon) models achieved the highest predictive accuracy, their intrinsic complexity and "black-box" nature limit interpretability and transparency—key factors for environmental decision-making. To address this, the SHapley Additive Explanations (SHAP) framework, grounded in game theory, was employed to quantify each feature's contribution to the predicted Water Quality Index (WQI) and to enhance the interpretability of both machine learning (ML) and deep learning (DL) models.

Global interpretability analysis

Figure 5 presents the importance of the SHAP-based global feature for SVR, Ridge, and DNN models across winter and monsoon seasons in the Padma River basin. SHAP values quantify both individual and interactive effects of input parameters, offering a consistent measure of feature influence on WQI predictions (Islam et al., 2025).

During winter, the SVR model identified pH and its interaction terms (pH–SO_4^{2-}, pH–Cl^-) as dominant predictors, alongside nitrate (NO_3^-) (Fig. 5a). The broad spread of SHAP values for these variables indicates strong nonlinear behavior, with pH-related parameters exhibiting both positive and negative impacts on WQI depending on their magnitude—reflecting complex acid–base and ionic balance dynamics in cooler, low-flow conditions (Nair et al., 2024).

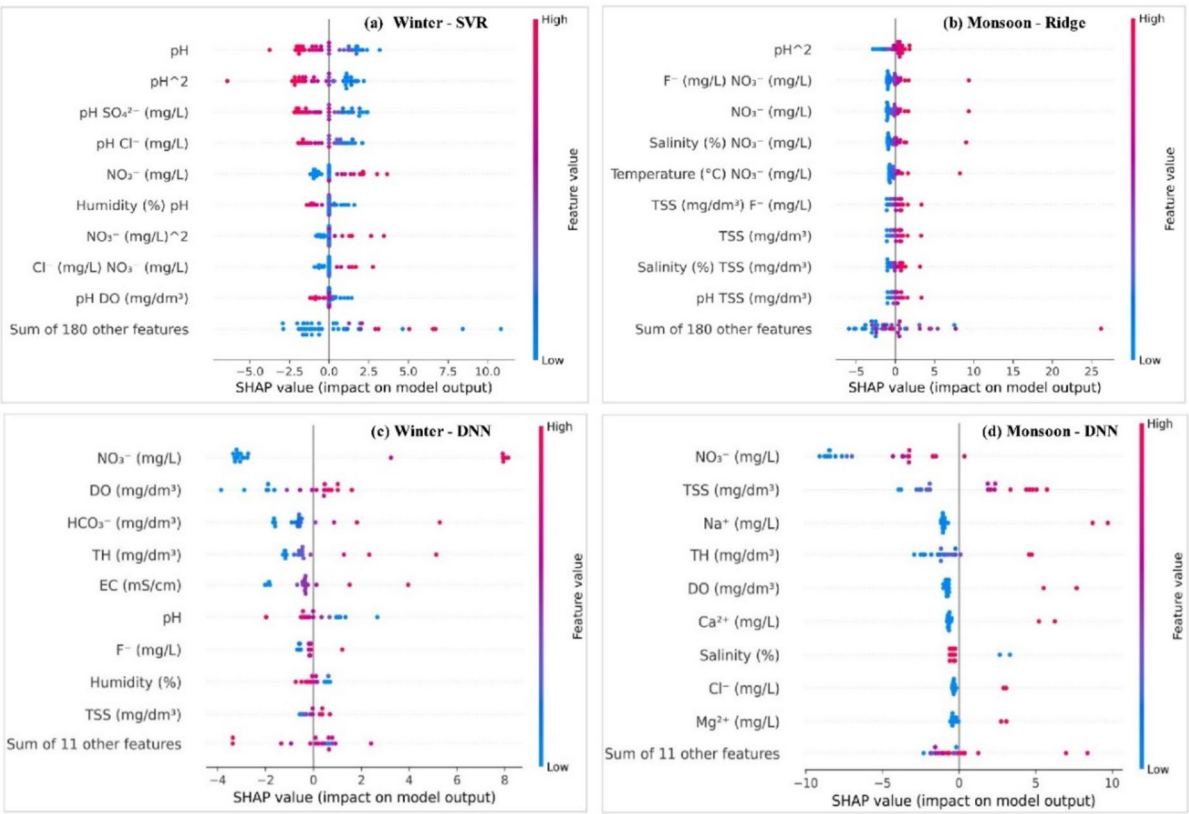


Fig. 5 SHAP-based global feature importance for WQI prediction using machine learning and deep learning models in the Padma River basin. **a** Winter–SVR, **b** Monsoon–Ridge, **c** Winter–DNN, and **d** Monsoon–DNN

The Ridge model showed a change in importance during the monsoon. While pH[2] and NO_3^- were still important, fluoride (F^-), salinity, temperature, and total suspended solids (TSS) became important as well (Fig. 5b). This change shows that more solutes are coming in from runoff and that water is mixing more during times of high flow.

The DNN models exhibited more complex attribution patterns. In winter, NO_3^-, DO, HCO_3^-, TH, and EC were the most influential features (Fig. 5c), indicating that ionic balance and oxygen dynamics play significant roles in determining WQI under relatively stable hydrological conditions. Conversely, in the monsoon, NO_3^-, TSS, Na^+, TH, and DO dominated (Fig. 5d), highlighting the interplay between sediment load, ionic composition, and oxygen availability during intensified flow regimes.

Color gradients in SHAP plots (red = high feature value; blue = low feature value) illustrate the direction and magnitude of influence. The variation in color intensity and spread demonstrates that feature impacts vary with season and model architecture.

Overall, the SHAP-based interpretability analysis reveals that feature importance is strongly season-dependent, governed by hydrological dynamics and model-specific sensitivities. These insights confirm that explainable AI (XAI) improves transparency and enhances confidence in model-based water quality assessments for complex, monsoon-driven river systems.

Local interpretability analysis

Figure 6 shows the local SHAP value analysis for WQI predictions in the Padma River basin. It shows how each water quality parameter affects the SVR, Ridge, and DNN model outputs during the winter and monsoon seasons. These results represent recent advances in interpretable machine learning, where SHAP values quantify and visualize the localized influence of each feature on predictive outcomes.

In winter, the SVR model identified nitrate (NO_3^-) as the most influential positive contributor to WQI, followed by temperature–nitrate and nitrate–sulfate interactions (Fig. 6a). In contrast, total hardness (TH) and dissolved oxygen (DO) exhibited negative SHAP values, signifying their inverse relationship with WQI under low-flow conditions. This finding aligns with other studies emphasizing the pivotal role of nutrient enrichment and oxygen depletion in shaping water quality indices during dry seasons (Yao et al., 2024).

The Ridge model showed an apparent change in feature contributions during the monsoon. Most parameters including nitrate, sulfate, and total suspended solids (TSS) displayed negative SHAP values, reducing WQI scores. In contrast, only the pH-fluoride interaction exerted a modest positive influence (Fig. 6b). This transition highlights how hydrological dilution and sediment influx during monsoon flows modify the relative importance of ionic and particulate variables.

For the DNN models, local interpretability also showed differences between seasons. In winter, DO was the dominant positive contributor. At the same time, nitrate exerted the most significant adverse effect on WQI (Fig. 6c). In contrast, during the monsoon, TSS became the main positive driver, while nitrate and TH caused predicted WQI values to drop significantly (Fig. 6d).

The local SHAP interpretations show how hydrochemical parameters that affect water quality interact in a complicated, nonlinear, and seasonal way. These insights emphasize that explainable AI (XAI) enhances transparency and provides valuable diagnostic understanding of parameter-specific effects in monsoon-influenced river systems. The results collectively affirm significant seasonality in feature importance influenced by hydrological dynamics and model-specific sensitivities, directly addressing Objective (iii) by offering a mechanistic, explainable-AI interpretation of how essential physicochemical parameters govern WQI predictions in a monsoon-dominated river system.

Probabilistic distribution and uncertainty analysis through monte Carlo simulation

Figure 7 compares the probabilistic distributions of WQI values predicted by various models—Ridge Regression, SVR, Voting Ensemble, and DNN—against actual and oversampled datasets across winter and monsoon seasons.

Figure 7a illustrates the probability density functions (PDFs) for predicted and observed WQI values. The actual WQI distribution exhibits a bimodal pattern with peaks near 25 and 60, reflecting seasonal contrasts and potentially distinct pollution regimes. The SVR model closely follows the lower peak

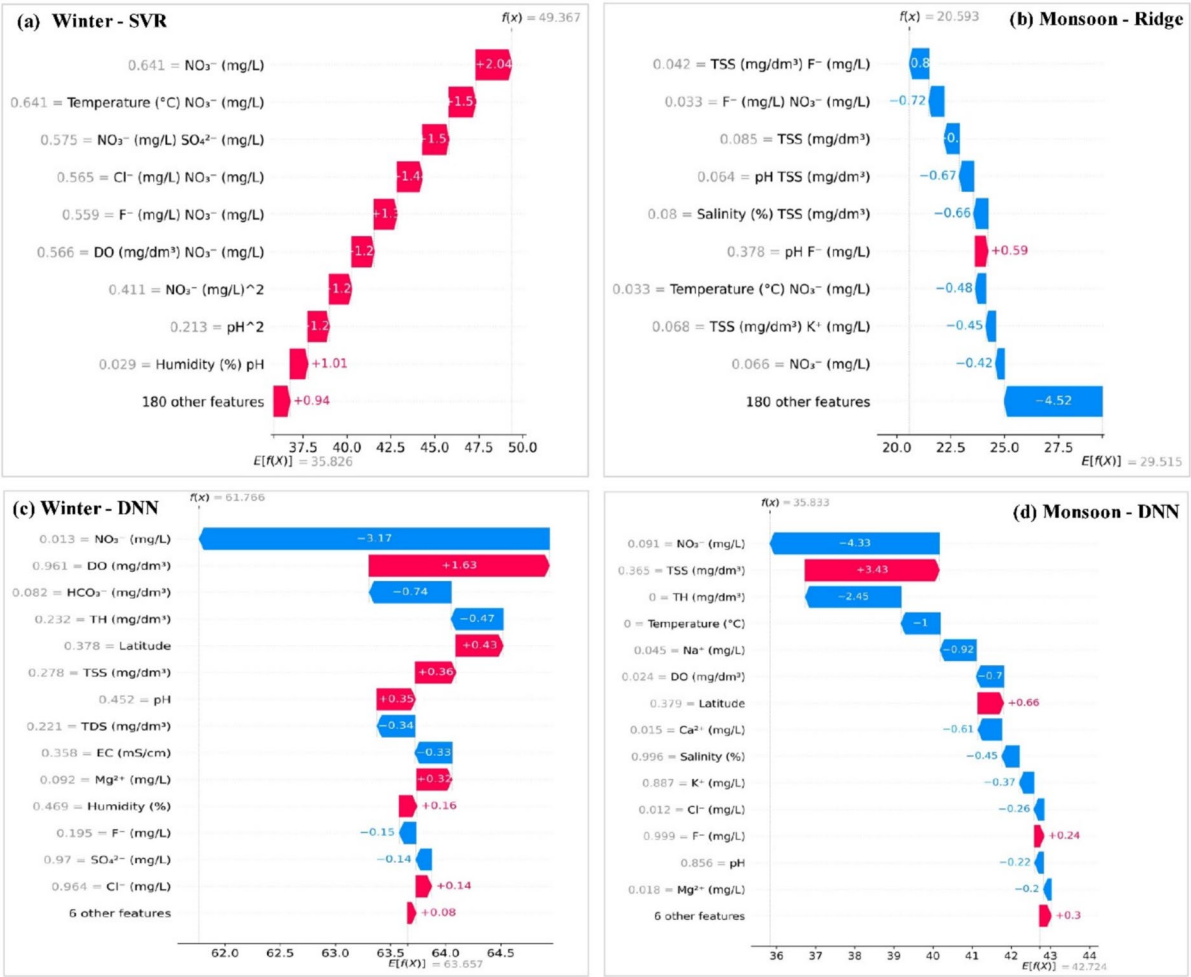


Fig. 6 Local SHAP explanations for WQI prediction in the Padma River basin using optimized machine learning and deep learning models. **a** Winter–SVR, **b** Monsoon–Ridge, **c** Winter–DNN, and **d** Monsoon–DNN. Waterfall plots show the contribution of individual features to model outputs, with red indicating positive and blue indicating negative SHAP impacts. The base value E[f(x)] represents the average model prediction

around 25, indicating a tendency to underpredict WQI and emphasize poorer water quality. In contrast, the Ridge and Voting Ensemble models capture both modes with smoother, less distinct peaks. The DNN and its oversampled variant display a sharp unimodal peak near 60, accompanied by a smaller secondary mode around 85, suggesting a bias toward higher WQI estimates and possible overprediction of water quality under certain conditions.

Figure 7b presents the corresponding cumulative distribution functions (CDFs), highlighting differences in spread and skewness among models. The SVR curve rises steeply at lower WQI values, reaffirming its consistent bias toward predicting degraded conditions. Conversely, the DNN and oversampled DNN models exhibit a delayed ascent, with much of their cumulative probability concentrated above WQI = 60, delineating the threshold between moderate and good quality classes. The shaded uncertainty zone (50–80% cumulative probability) represents the range where classification ambiguity is most significant. Within this interval, the Voting Ensemble CDF aligns most closely with the actual data, indicating superior reliability in reproducing the observed variability and uncertainty of water quality conditions.

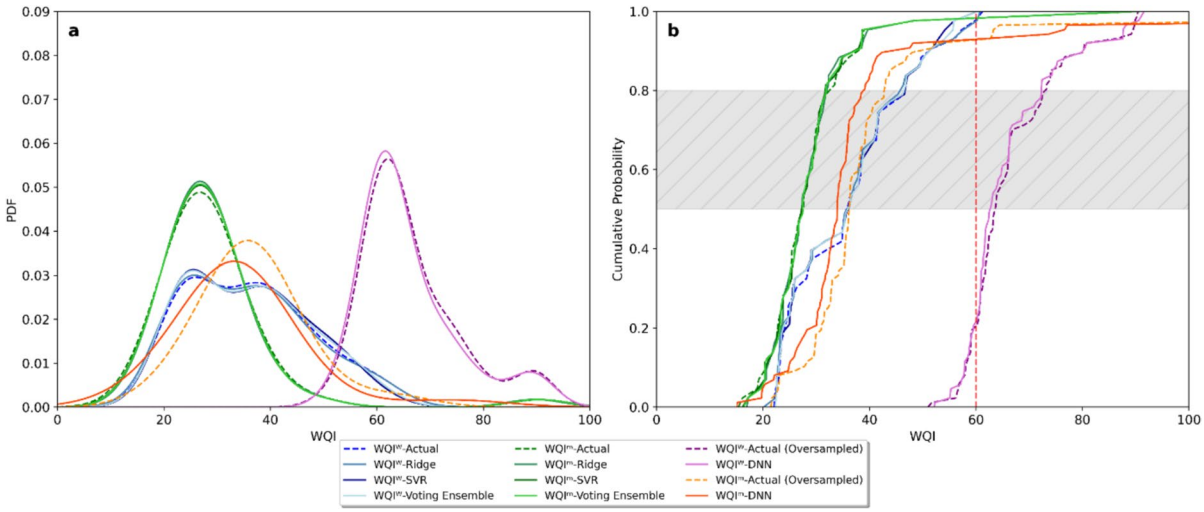


Fig. 7 Probability and cumulative distribution plots of predicted WQI values using machine learning and deep learning models during winter and monsoon. **a** Probability density functions (PDFs) of predicted and actual WQI values, including oversampled models. **b** Corresponding cumulative distribution functions (CDFs) showing distributional spread and skewness across modeling strategies. Dashed lines represent actual WQI distributions, and solid lines represent model predictions. Shaded band (50–80%) and WQI threshold (60) highlight classification zones and uncertainty range

The Monte Carlo simulation results demonstrate that while deep learning models achieve high accuracy, they may overfit higher-quality regimes. In contrast, ensemble approaches better capture the probabilistic structure and uncertainty of real-world water quality distributions across seasons.

Figure 8 shows the likely distributions of WQI values in the Padma River basin for the winter and monsoon seasons. This is based on 20,000 Monte Carlo simulation trials for each season. Both distributions exhibit right-skewed, bell-shaped patterns, indicating that most WQI values cluster around moderate levels, with a few instances of higher values representing less frequent but possible episodes of poorer water quality.

In winter, simulated WQI values range from approximately 12 to 52, with the 25th and 75th percentiles at 20.00 and 42.20, respectively. The mean WQI (30.85) exceeds the median (26.43), suggesting a slight positive skew toward higher values. The distribution's peak probability occurs between 24 and 32, indicating that most simulated outcomes fall within this range.

The WQI distribution moves to the left during the monsoon, covering a range of about 8 to 56. The 25th and 75th percentiles are 14.51 and 40.53, respectively. The mean (27.14) and median (23.13) are both lower than in winter, signifying improved water quality under enhanced river flow conditions. The peak interval (20–28) further reflects this seasonal improvement due to increased dilution and reduced pollutant concentrations.

Both seasonal distributions align closely with lognormal fit curves, confirming an excellent statistical representation of simulated data. Vertical reference lines for the mean, median, and percentile thresholds make it easy to compare seasons side by side. The overall leftward shift and lower central tendency of monsoon WQI values reinforce the conclusion that higher discharge during the rainy season enhances dilution capacity, leading to better water quality, whereas restricted flow in winter favors solute accumulation and pollutant persistence. This probabilistic characterization enhances the deterministic modeling outcomes by explicitly quantifying seasonal uncertainty and degradation risk in WQI, hence bolstering confidence in model-based classification under fluctuating hydrological conditions.

Seasonal spatial distribution and seasonal classification of WQI

Figure 9 illustrates the spatial distribution of WQI values across the Padma River basin during winter

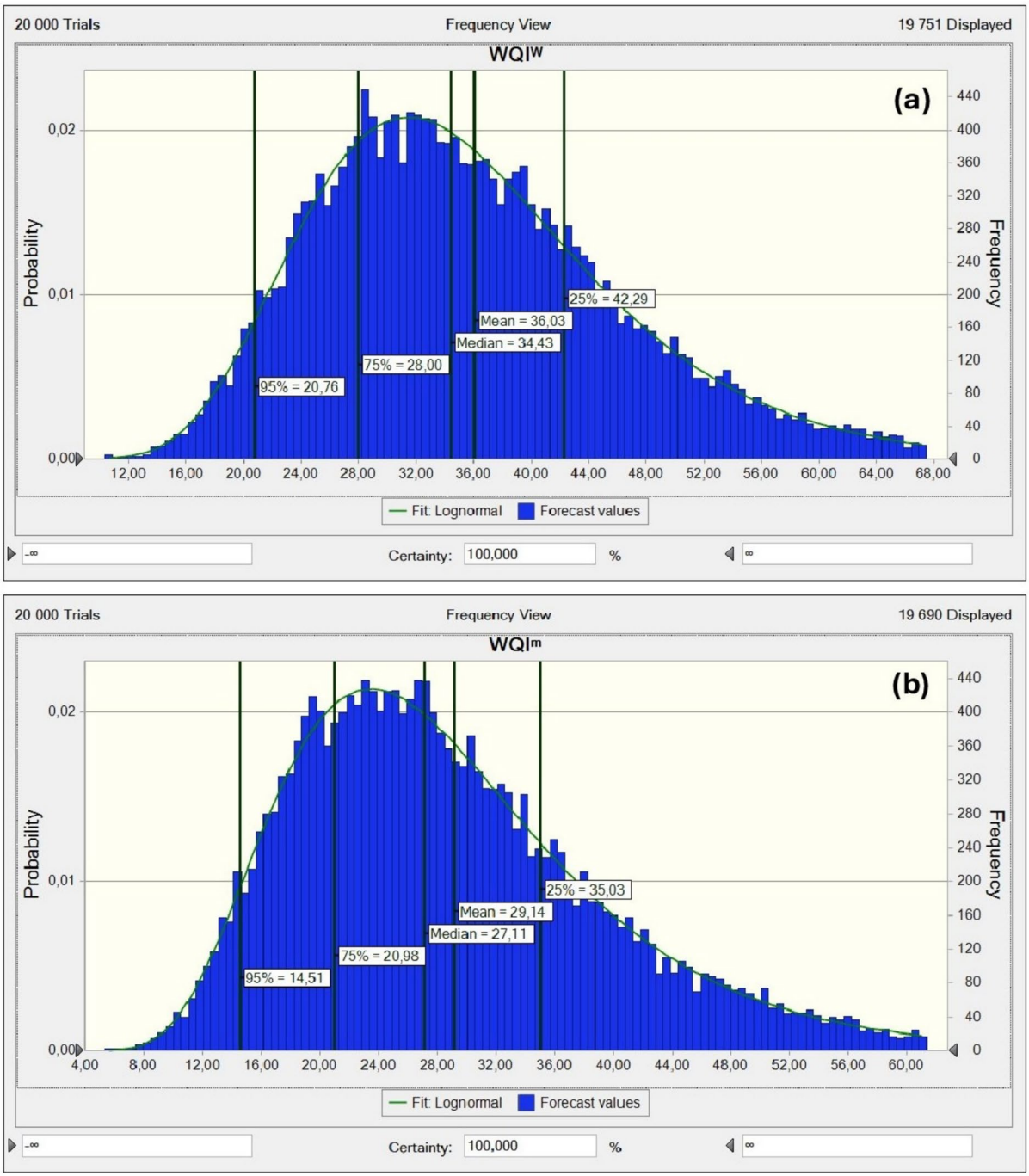


Fig. 8 Monte Carlo simulation-based probabilistic distribution of WQI across winter and monsoon seasons in the Padma River basin. Panels **a** and **b** show frequency histograms (blue bars) from 20,000 simulation trials with lognormal distribution fits (black curves) for winter and monsoon, respectively. Vertical black lines mark key percentiles (5%, 25%, median, 75%, 95%), and white boxes display statistical summaries quantifying prediction uncertainty ranges

and monsoon seasons, revealing pronounced seasonal contrasts.

During winter, WQI values exhibit considerable spatial variability. The upstream and central reaches predominantly fall within the "Good" to "Poor" categories, while a localized mid-basin segment (sites S7–S14) records "Poor" to "Very Poor" quality, reflecting elevated pollution and reduced dilution capacity during the dry season. Conversely, the downstream reaches (sites S37–S44) consistently display "Excellent" water quality, likely due to increased flow volume, dilution effects, and lower anthropogenic pressure (Table S11 and Fig. S10).

In contrast, monsoon conditions markedly improve water quality across the basin. The entire river corridor transitions to "Excellent" or "Good" classes, with no areas classified as "Poor," "Very Poor," or "Unsuitable for drinking." Notably, previously degraded midstream zones (S7–S14) improve to "Good" or "Excellent," emphasizing the dominant role of hydrological variability—particularly monsoonal flushing and dilution—in shaping water quality patterns (Table S11 and Fig. S11).

Figure 10 presents a Sankey diagram depicting seasonal transitions in WQI classes (Excellent, Good, and Poor) from winter to monsoon. The majority of sites maintain "Good" status year-round, as shown by the dominant green stream, while several "Good" sites shift to "Excellent" during the monsoon, indicating seasonal improvement. A smaller fraction of "Poor" winter sites transition to "Good" or "Excellent," demonstrating recoverability under high-flow conditions. Only a few flows enter the "Unsuitable for drinking" category, suggesting that extreme degradation events are rare. Overall, the diagram highlights the persistence and variability of water quality classes, reflecting the dynamic equilibrium of the Padma River system.

Figure 11 further integrates these findings by mapping the spatial-seasonal dynamics of WQI classifications across 44 sampling sites. In winter (Fig. 11a), 62% of sites fall within "Good" (Class II), 27% as "Excellent" (Class I), and 11% as "Poor" (Class III). Spatial connectivity lines indicate clustering similar WQI zones, suggesting localized pollution sources and persistent hydrochemical regimes.

In monsoon (Fig. 11b), 98% of sites shift to "Excellent" or "Good," with only 2% remaining "Poor."

The comparative classification matrix (Fig. 11c) highlights both stable and variable sites, showing that stations degraded in winter frequently improve by one or two classes during monsoon flow conditions.

These spatial and temporal analyses confirm that hydrological seasonality, particularly the monsoonal discharge regime, is the principal driver of water quality variability in the Padma River basin, enhancing dilution and flushing during wet periods and concentrating pollutants during the dry season. The identified geographical clustering patterns and seasonal

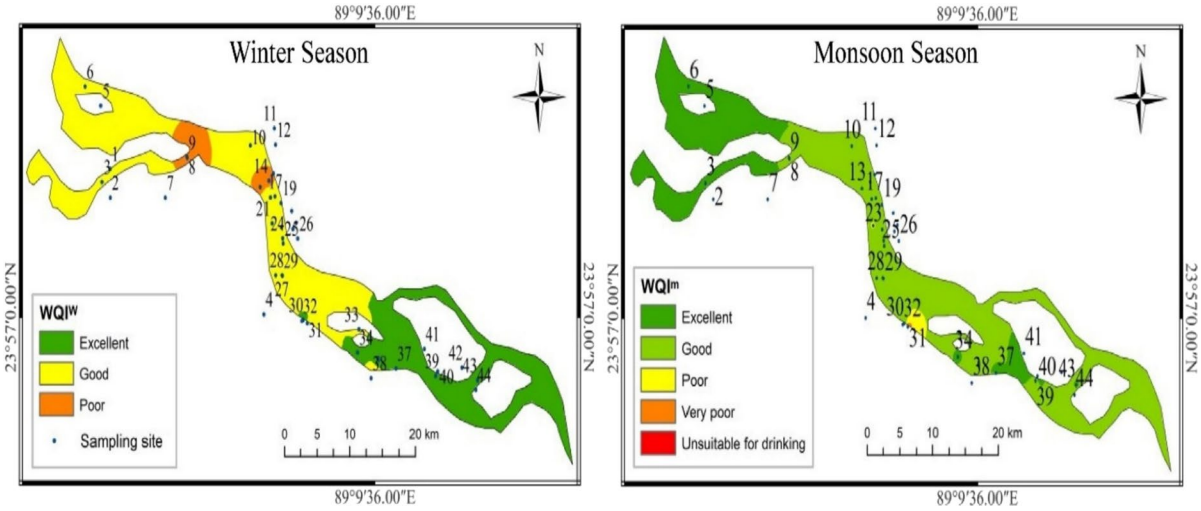


Fig. 9 Spatial distribution and seasonal classification of WQI values in the Padma River basin during winter and monsoon

Fig. 10 Sankey diagram illustrating seasonal transitions in WQI classes from winter to monsoon across the Padma River basin

class changes validate the study hypothesis that basin-scale water quality dynamics are predominantly influenced by hydrological forces rather than enduring spatial controls.

Spatial regionalization of seasonal WQI in the Padma River basin

Figure 12 presents the spatial regionalization and clustering patterns of WQI values across the Padma River basin, based on Local Indicators of Spatial Association (LISA) and Moran's I analysis, highlighting how water quality varies spatially between winter and monsoon seasons.

During the winter season (Fig. 12a), the LISA cluster map reveals pronounced spatial heterogeneity. Approximately 34.1% of sites form high–high clusters—zones where both a site and its neighbors exhibit elevated WQI values—concentrated primarily in the northwestern basin. Conversely, low–low clusters (20.5%) occur mainly in the eastern sector, representing contiguous areas of degraded water quality. Spatial outliers, such as high–low (13.6%) and low–high (31.8%) clusters, show localized anomalies or transitional zones where the water quality differs greatly from nearby sites. This could be due to point-source pollution or hydrochemical mixing zones.

In the monsoon season (Fig. 12b), spatial clustering patterns shift markedly. High–high clusters drop to 20.4% and spread out more, while low–low clusters grow to 27.3%. This suggests that larger areas of localized degradation are probably caused by runoff and sediment resuspension. The proportions of low–high (31.8%) and high–low (20.5%) clusters remain comparable but exhibit increased dynamism, indicating that monsoonal hydrology enhances spatial variability and redistributes pollutants across neighboring sites.

Figure 12c shows the Moran scatterplots for both seasons, plotting local Moran's I value against WQI scores. The upper-right quadrant shows high–high clusters (WQI > 40), which means strong positive spatial autocorrelation among higher-quality zones. The lower-left quadrant shows low–low clusters (WQI < 30), meaning there is negative spatial autocorrelation in degraded areas. Vertical reference lines at WQI ≈ 30 and 40 delineate the thresholds separating poor and excellent quality classes, reinforcing the spatial relevance of these limits.

These results demonstrate that the Padma River basin exhibits significant spatial clustering of water quality, with the extent, intensity, and spatial positioning of high- and low-quality zones shifting seasonally in response to hydrological variability and land-use pressures. The findings draw attention to the importance of seasonally adaptive and spatially targeted management interventions to address persistent contamination hotspots and emerging cold spots for improving water quality. It should be noted that the identified LISA and Moran's I patterns represent snapshot spatial organization under contrasting seasonal flow regimes in a single year, and should not be interpreted as persistent or inter-annually stable spatial clusters.

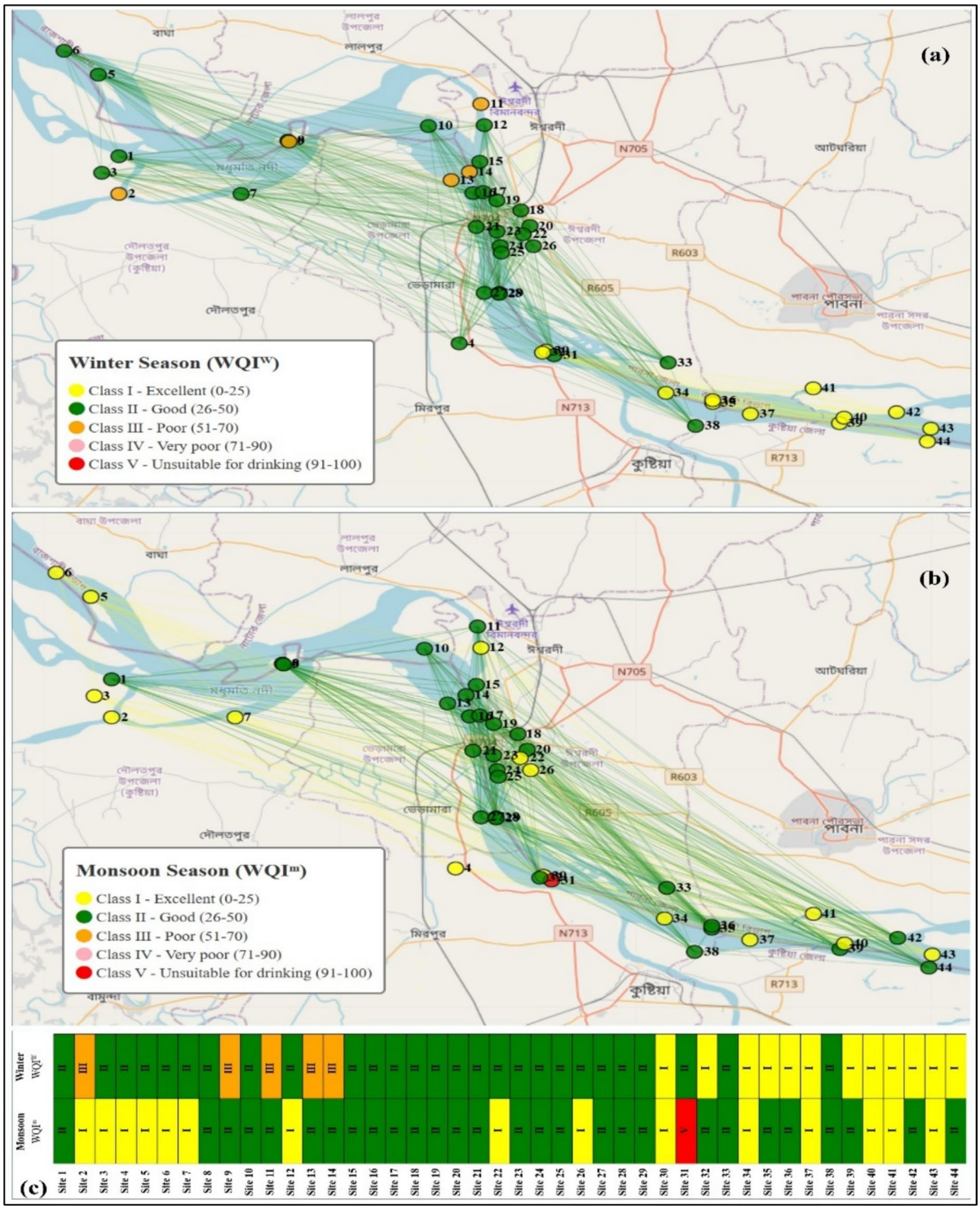


Fig. 11 Seasonal classification of WQI across 44 sampling sites with spatial network connectivity of similarly classified zones in the Padma River basin. **a** WQI classes during winter, **b** WQI classes during monsoon, and **c** comparative matrix of seasonal classification

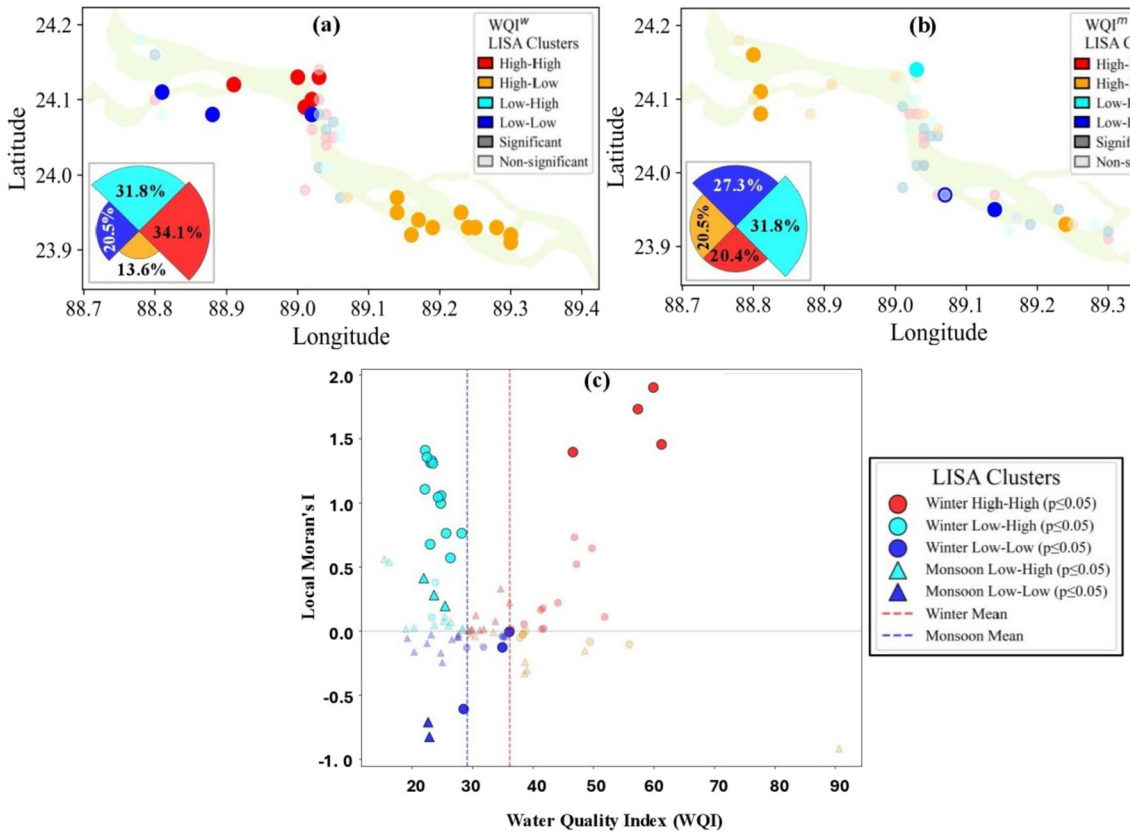


Fig. 12 Local indicators of spatial association (LISA) and Moran's I analysis of seasonal WQI in the Padma River basin. **a** Winter-season spatial clusters of WQI, highlighting high–high, low–low, and other significant associations. **b** Monsoon-season clustering, showing seasonal shifts in hotspot and cold spot zones. **c** Moran scatterplot depicting local spatial autocorrelation of WQI for both seasons. Cluster categories are defined as high–high, high–low, low–high, and low–low, based on significance at $p < 0.05$

Discussion

Hydrological and land-use controls on seasonal water quality variability

The seasonal water quality variations in the Padma River are best understood as a result of interconnected hydrological influences, catchment processes, and in-channel dynamics, rather than as independent parameter behaviors. Increased electrical conductivity, total dissolved solids, hardness, and major ions during winter correlate with diminished river discharge, restricted dilution capacity, and solute accumulation under low-flow conditions, a trend commonly observed in extensive monsoon-influenced river systems (Islam et al., 2024; Uddin et al., 2017, 2018). The seasonal abstraction for irrigation and municipal purposes during the dry season may exacerbate these concentration effects, however the investigation did not specifically quantify the abstraction rates.

The systematic decrease of dissolved elements during the monsoon indicates rainfall-induced dilution and enhanced longitudinal connection, resulting in the homogenization of water chemistry at the basin scale (Bui et al., 2020; Paneru & Paneru, 2025). The significant rise in total suspended solids and simultaneous decrease in dissolved oxygen during high-flow times signify increased sediment mobilization, organic matter transport, and oxygen use. These responses align with watershed erosion, floodplain inundation, and in-channel resuspension processes typically observed in alluvial rivers experiencing severe monsoon rains (Bishwakarma et al., 2025; Deng et al., 2022).

This study highlights hydrologically mediated correlations rather than direct causal relationships. Increased suspended solids during the monsoon correlate with upstream land-use disturbances and agricultural activities; however, the existing data do not provide the differentiation of contributions from hillslope erosion, bank erosion, and in-channel resuspension. Nitrate variability during the monsoon corresponds with diffuse non-point source mobilization from agricultural land; however, fertilizer inputs cannot be differentiated from other nitrogen sources without isotopic or source-apportionment analysis. The results should be seen as evidence of process coherence with the hydrological regime rather than as proof of specific source causation.

Model performance and process-consistent interpretation

The seasonally variable efficacy of prediction models indicates variations in hydrochemical stability among hydrological regimes, rather than the inherent superiority of specific algorithms. The superior winter performance of the deep neural network indicates that nonlinear relationships among physicochemical parameters are more stable in low-flow conditions, where variability is primarily influenced by concentration processes and localized inputs (Khosravi et al., 2025; Zheng et al., 2024). In the monsoon season, heightened variability, noise, and nonstationarity diminish the efficacy of highly flexible models, hence favoring ensemble and regularized methods that more effectively address hydrological changes (Hui et al., 2025; Nair et al., 2024).

Explainable AI analysis offers further understanding of these regime-specific controls by detecting transitions in primary predictors from ionic and oxygen-related factors in winter to sediment- and nutrient-related factors during the monsoon. These results align with theoretical frameworks of monsoon river dynamics and endorse the application of XAI as a diagnostic instrument for elucidating data-driven forecasts (Nong et al., 2025; Satish et al., 2024). Nonetheless, SHAP-derived significance values measure statistical contributions to model output rather than physical causality, and should thus be regarded as markers of predominant relationships rather than conclusive mechanistic causes.

Comparison with similar river studies and methods

The dominance of seasonal hydrological forcing observed in this study aligns with findings from other large, monsoon-influenced river systems across South and Southeast Asia (Table S1). Studies in Himalayan and transboundary rivers have similarly reported winter concentration regimes and monsoon dilution accompanied by sediment and nutrient mobilization (Bishwakarma et al., 2025; Paneru & Paneru, 2025). Compared with studies relying solely on traditional WQI trend analysis or process-based watershed models such as SWAT, the present framework offers complementary insight by explicitly integrating explainable prediction, uncertainty characterization, and spatial classification.

International applications of machine learning for river water quality prediction increasingly emphasize the need for interpretability and uncertainty awareness to support decision-making (Zheng et al., 2024; Priyadharshini & Ramesh, 2025). In this context, the present study advances existing approaches by demonstrating how explainable AI and probabilistic analysis can be combined to reveal regime-specific controls and spatial risk patterns, particularly in data-limited settings where full process-based modeling may not be feasible. Nevertheless, comparable studies in monsoon-driven river systems support the robustness of strong seasonal forcing despite similar temporal constraints. For example, Nguyen et al. (2022) reported 88% correct seasonal classification between dry and rainy periods using discriminant analysis across 49 samples in the Vietnamese Mekong Delta coastal region. Similarly, Ali et al. (2024) found significant seasonal variation in BOD_5 (t-test, $p < 0.05$) but non-significant spatial variation among sampling stations (one-way ANOVA, $p > 0.05$) in Bangladesh's Dhaleshwari River. Kabir et al. (2021) further corroborated this pattern in the Shitalakhya River, Bangladesh, reporting statistically significant seasonal variation (ANOVA, $p < 0.05$) for 13 of 14 water quality parameters (excluding SO_4^{2-}). In fact, these regional studies emphasise confidence that the seasonal patterns observed in this study reflect characteristic subtropical river dynamics, while underscoring the need for multi-year validation to establish interannual stability.

Model uncertainty

Despite strong performance, predictive models inherently carry structural, parametric, and data-related uncertainties. Deep Neural Networks (DNNs) are highly effective under winter conditions ($R^2 = 0.98$). However, they may show reduced alignment with monsoon dynamics ($R^2 = 0.88$). This change can result from overfitting risks or shifts in input data distributions. The limited sample size (n = 44) further amplifies parametric uncertainty. Measurement errors introduce minimal bias (RSD < 5%). Common challenges in environmental modelling, such as missing data (Pan et al., 2025), were minimized through rigorous quality assurance and quality control (QA/QC) procedures (Table S12).

The reliance on PINN-generated synthetic samples warrants careful interpretation. However, several independent validation phases mitigate concerns regarding overfitting to model-generated patterns: (1) model evaluation on test datasets composed exclusively of real, held-out samples rather than PINN-generated data; (2) learning-curve analyses demonstrating convergence between training and validation errors without divergence (Fig. S12); (3) physics-based loss functions enforcing hydrochemical constraints, including charge balance (< 10%), mass balance, and ratio preservation; and (4) cross-validation stability of ensemble models (Table S10). Recent studies employing comparable augmentation strategies (Priyadharshini et al., 2025; Rahaman et al., 2024) further demonstrate that physics-informed synthetic data generation can preserve predictive validity when appropriately constrained. The observed reduction in DNN performance from winter to monsoon conditions ($R^2 = 0.98$ to 0.88), therefore, reflects the substantially higher variability of monsoon hydrochemical parameters, such as dissolved oxygen (CV = 343% vs. 7.7%; Tables S8–S9), rather than systematic overfitting. Consistently, ridge regression outperformed DNNs during the monsoon season ($R^2 = 0.96$), indicating that linear, regularized models are more robust to high-variance and noisy data than nonlinear architectures under such conditions.

Importantly, using single-year data (2024) limits the ability to distinguish persistent seasonal patterns from interannual variability. ANOVA results indicate highly significant seasonal effects ($F > 100$, $p < 0.0001$). These findings reflect a single hydrological cycle, which may not capture long-term climate oscillations, or progressive anthropogenic influences.

Implications for river management, policy, and future monitoring

The results yield distinct, policy-relevant insights for river management in monsoon-dominated, data-scarce basins. The pronounced seasonal variability in water quality suggests that annual average evaluations are inadequate and may conceal times of heightened ecological and human health risks (Islam et al., 2024). Thus, regulatory assessments and monitoring initiatives should explicitly differentiate between low-flow (dry season) and high-flow (monsoon) conditions, with special emphasis on winter periods as critical phases of degradation (Jawad-ul-Haque et al., 2023). The recognition of nitrate- and sediment-related parameters as regime-specific stressors indicates that management strategies must be seasonally tailored, focusing on point-source control, abstraction management, and oxygen stress alleviation in the dry season, while prioritizing non-point source reduction, erosion control, and floodplain management during the monsoon (Kabir et al., 2021). The current analysis is unable to assess the efficacy of specific upstream interventions, land-use policies, or flow regulation measures, nor can it attribute accountability to particular sectors or facilities. Such determinations necessitate the integration of operational data, regulatory documentation, and source-tracing methodologies (Hu et al., 2023). The results should be regarded as decision-support intelligence rather than prescriptive regulation. Future initiatives should prioritize the expansion of monitoring networks, enhancement of temporal resolution, and integration of interpretable data-driven models with hydrological and land-use datasets to progress from pattern recognition to causal inference and adaptive river basin management alignment with SDG 6 (Clean Water and Sanitation). The technique enhances interpretability using XAI, enabling water managers to pinpoint key pollution drivers, e.g., nitrates and suspended solids, thereby improving resource allocation and promoting climate-resilient management strategies in response to changing hydrological conditions. The method provides a scalable, evidence-based framework for sustainable

water governance in monsoon-affected river systems in Bangladesh and similar regions worldwide.

Study limitations and future research needs

This research provides a comprehensive framework for forecasting seasonal river water quality. However, several limitations must be acknowledged that the current sampling density cannot resolve fine-scale spatial heterogeneity, inter-annual variability remains unaddressed, and validation of persistent seasonal variability and spatial clustering requires higher-frequency sampling. The empirical dataset comprises 44 grab samples collected during one winter and one monsoon season in 2024, without interannual replication and neglecting lateral and tributary effects. As a result, statistical analyses such as ANOVA, LISA, and Moran's I are inherently sensitive to sampling density, year-specific hydrological conditions, and short-term anomalies. Consequently, the detected seasonal contrasts and spatial structures should be interpreted as diagnostic patterns within the sampled year, rather than definitive evidence of long-term or basin-wide behavior. Separating persistent seasonal signals from interannual variability will require multiyear monitoring with higher spatial density.

The PINN-based data augmentation facilitated the training of deep learning models with low sampling while maintaining the limitations of hydrochemical data; nonetheless, almost 90% of the training data were supplemented rather than directly observed. Subsequent monitoring period is essential to verify the model's transferability beyond the 2024 Padma River dataset.

This research focused on physicochemical factors and excluded biological markers, which might provide further insights into ecosystem health and deserve future ecological assessments. Although scenario-based studies were not conducted, the proposed framework enables future evaluations of climatic variability, land-use changes, and policy interventions via regulated parameter adjustments within physics-informed limitations. A comprehensive scenario-based study will be enhanced by multi-year baseline data, stakeholder-informed assumptions, and extensive RNPP operating information in the future. Despite these drawbacks, the study's statistical robustness (median $\eta^2 = 0.61$; power > 0.95 for 88% of parameters), alignment with similar monsoon-driven river basin studies (Ali et al., 2024; Kabir et al., 2021; Nguyen et al., 2022). Physics-informed modeling methodology validates the rationality of the primary conclusions concerning seasonal predominance, spatial clustering, and model efficacy.

Conclusion

This study demonstrates that seasonal hydrological forcing predominantly governs river water quality in the monsoon-influenced Padma River Basin. Winter low-flow conditions promote solute accumulation and localized water quality degradation, whereas monsoon discharge drives basin-wide dilution and recovery. Model performance varies systematically with hydrological regime: deep neural networks achieve optimal accuracy under relatively stable winter conditions, while regularized and ensemble models exhibit greater robustness during the highly variable monsoon season. These findings underscore the necessity of season-adaptive modeling strategies for reliable water quality prediction. Explainable AI analysis identifies nitrate as the dominant driver of Water Quality Index variability in both seasons, while pH, dissolved oxygen, suspended solids, and hardness exert nonlinear, season-dependent influences, reflecting consistent hydrochemical processes rather than spurious correlations. The integration of physics-informed data augmentation improves model generalization under limited sampling conditions, and combined uncertainty-aware and spatial analyses reveal coherent seasonal risk patterns characterized by localized winter degradation hotspots and widespread monsoon improvement. Overall, the proposed framework provides a transparent, interpretable, and uncertainty-aware decision-support tool for understanding intra-annual water quality dynamics in data-scarce, monsoon-driven river systems. Its primary contribution lies in methodological integration rather than long-term trend attribution. Future research should prioritize high-frequency and event-based monitoring, incorporate hydrological and land-use drivers to strengthen causal attribution, further refine physics–AI hybrid models under nonstationary conditions, and enhance integration with management and stakeholder frameworks to support adaptive, uncertainty-aware decision-making in monsoon-affected river basins.

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Author Contribution ABMTI., MNU, TRC, MHC, NEA, SFFS and SRK wrote the main manuscript text, and M.A.A.M. prepared Figs. 1–9. All authors reviewed the manuscript.

Declarations

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the writing process During the preparation of this work the authors used Grammarly and ChatGPT in order to improve the readability and language of the manuscript. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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